

GENUS MONOTONICITY FOR NUMERICAL SEMIGROUPS

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ABSTRACT. Let n_g denote the number of numerical semigroups of genus g . We prove that $n_{g+1} \geq n_g$ for every $g \geq 0$, settling the weak monotonicity form of the Bras-Amorós conjecture. The computer-assisted proof constructs an injection outside a specified family of leaves and bounds this exceptional family by a collection of unused targets. Generating-function estimates establish the comparison for $g \geq 837$; exact transfer computations bound relaxed counts of the exceptional sources for $1 \leq g \leq 836$. The analytic estimates distinguish genuine semigroups from a larger two-pivot relaxation, whose coefficients can be computed by transfer along paths and cycles.

1. INTRODUCTION

A numerical semigroup is an additive submonoid S of $\mathbb{N}_0$ with finite complement. Its genus is $g(S) = |\mathbb{N}_0 \setminus S|$. Write

$$\mathcal{S}_g = \{S : g(S) = g\}, \quad n_g = |\mathcal{S}_g|.$$

Bras-Amorós's computations led her to conjecture the stronger Fibonacci inequality for this sequence [2]. We consider its weak monotonicity form, recorded separately in Kaplan's survey [7, Conjecture 2].

Theorem 1.1. *For every integer $g \geq 0$, one has $n_{g+1} \geq n_g$.*

The conjecture concerns all numerical semigroups, with no restriction on multiplicity or conductor. It is weaker than the Fibonacci inequality $n_g \geq n_{g-1} + n_{g-2}$ for $g \geq 2$, which is not established here. The distinction between all genera and sufficiently large genus is essential. Zhai proved that

$$(1.1) \quad n_g \sim C\varphi^g, \quad \varphi = \frac{1 + \sqrt{5}}{2}, \quad C > 0.$$

Consequently $n_{g+1}/n_g \rightarrow \varphi > 1$, so eventual strict increase was already known [13]. An all-genus proof requires control of the range preceding that asymptotic conclusion. Exact values of n_g have been computed through genus 80 [3, Section 4].

The argument starts with an incomplete injection rather than an estimate of n_g itself. For each g , we construct an injection from $\mathcal{S}_g \setminus \mathcal{E}_g$ into $\mathcal{S}_{g+1}$ and exhibit at least $L(g)$ targets outside its image. Thus

$$(1.2) \quad n_{g+1} - n_g \geq L(g) - |\mathcal{E}_g|.$$

The exceptional sources are leaves in the numerical-semigroup tree and have depth four or five. The construction is arranged so that their remaining possibilities admit two different estimates: a coefficient bound for a finite range and a weighted generating-function bound for large genus. Neither estimate is required to approximate n_g closely.

The proof uses established low-depth constructions. Eliahou and Fromentin constructed two disjoint injections on gapsets of depth at most three, which become appending 1 and appending 2 in Kunz coordinates [6, Propositions 5.2–5.3]. Their results imply a Fibonacci lower recurrence for the low-depth count. Kunz coordinates and their interpretation as words or compositions

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also underlie the asymptotic work of Bacher and Zhu [1, 14]. Bacher uses maximal pivots, reflections and transfer matrices; Zhu proves sub-Fibonacci exponential bounds for the higher-depth population using weighted graph-homomorphism estimates. The general use of these methods is therefore established. Here they are combined with a specified exceptional family, disjoint unused targets, and explicit constants that permit the finite and analytic ranges to meet.

The all-genus monotonicity problem is distinguished explicitly from its stronger variants in recent work of Cyrusian and Kaplan [5]. Bras-Amorós and Torra obtain refined recurrences under multiplicity and Frobenius-number hypotheses [3]; Casas, Madrid and Rosales study related internal and leaf counts [4]. For semigroups of ordinarization number three, Kumar’s preprint combines a shift on a large class with a rank matching into a disjoint reserve, using separate finite and eventual capacity bounds [9, Sections 7–8]. This gives a restricted precedent for the allocation strategy used here. An earlier proof of monotonicity was proposed by Kachouei and Rahmati [8]. The proof of their Theorem 19 (p. 13) asserts

$$\binom{n_{m,g-1}}{m-1} \leq n_{m,g} \quad (n_{m,g-1} \geq m),$$

where $n_{m,g}$ counts semigroups of multiplicity m and genus g . However, $n_{3,9} = n_{3,10} = 4$, so the bound would require $6 = \binom{4}{2} \leq 4$. Indeed, multiplicity-three Kunz pairs satisfy $a + b = g$, $2a \geq b$ and $2b + 1 \geq a$, giving these counts directly. This concerns the proposed counting argument, not the monotonicity statement itself.

Our finite computation evaluates a relaxed counting problem and checks the inequalities in (1.2). It does not enumerate all numerical semigroups through genus 836, or compute all the corresponding values of n_g . We give the coverage and arithmetic justification separately from the algorithms’ implementation. The proof of Theorem 1.1 is completed after the finite and analytic comparisons have been established.

The population notation used throughout the proof is collected in Table 1. Auxiliary matrix entries and scalar bounds are local to the subsection in which they are defined.

TABLE 1. The principal populations and counting bounds.

Notation	Meaning
$\mathcal{S}_g, n_g$	All genus- g semigroups and their number
$\mathcal{E}_g$	Sources left unmatched by the partial injection
$L(g)$	Explicit lower bound on unused genus- $(g+1)$ targets
A_g, B_g	Genuine words with at least two maxima, of depth four and five
$A(z), B(z)$	Generating series of these two genuine populations
$Z_{m,r,p}(z)$	Raw depth-four partition with rightmost fours at $r < p$
$F_{r,p}(z)$	Sum of $Z_{m,r,p}(z)$ over $m > p$
U_g, V_g	Integer upper bounds used in the finite comparison

2. A PARTIAL INJECTION AND ITS EXCEPTIONAL SOURCES

We use positive Kunz coordinates [10, 12]. If a numerical semigroup S has positive genus and multiplicity m , let $i + mx_i$ be its least element congruent to i modulo m , for $1 \leq i < m$. Its Kunz word is $x = (x_1, \dots, x_{m-1})$. The empty word represents $\mathbb{Z}_{\geq 0}$. The weight of a word is $|x| = \sum_i x_i$; its depth is $\max_i x_i$ for a nonempty word. The empty word has weight and depth zero. A word is called binary if all its letters belong to $\{1, 2\}$. Throughout, coefficients with negative indices are zero.

Lemma 2.1. *Positive words of length $m - 1$ represent numerical semigroups of multiplicity m precisely when*

$$(2.1) \quad x_i + x_j \geq x_{i+j} \quad (i + j < m),$$

$$(2.2) \quad x_i + x_j + 1 \geq x_{i+j-m} \quad (i + j > m),$$

including self-pairs. The genus is $|x|$. If the depth is a and p is the rightmost coordinate equal to a , the Frobenius number is $(a - 1)m + p$.

Proof. In residue i , the gaps are $i, i + m, \dots, i + (x_i - 1)m$, proving the genus formula. Adding the least nongaps in two nonzero residues proves necessity of the displayed inequalities. Conversely, these inequalities ensure that sums of these least nongaps belong to the prescribed residue tails. Adding multiples of m then proves closure for all nongaps. A sum in residue zero already belongs to the semigroup, so no condition is needed when $i + j = m$. The largest gap in residue i is $i + (x_i - 1)m$, and comparison of these numbers proves the Frobenius formula. $\square$

We set the Frobenius number of $\mathbb{Z}_{\geq 0}$ to -1 . An *effective generator* is a minimal generator exceeding the Frobenius number. Deleting one gives an ordinary child of genus one greater. A semigroup without effective generators is a *leaf*. We call depth at most three low depth and depth at least four high depth.

Lemma 2.2. *Every positive-genus semigroup has a unique ordinary parent, obtained by restoring its Frobenius number. Every parent has at most one leaf child. For a high-depth word, restoring the Frobenius decreases its rightmost maximal coordinate by one and preserves multiplicity. Consequently a high-depth leaf has a low-depth parent exactly when its depth is four and it has a unique coordinate equal to four.*

Proof. Let F be the Frobenius number. Restoring F preserves closure: every sum of F and a positive nongap is larger than F , as is $2F$. In the enlarged semigroup, F cannot be a sum of two positive elements, for such a decomposition would already have contradicted closure before restoration. The new Frobenius is smaller than F , so F is an effective generator. Conversely, deletion of an effective generator a gives a semigroup whose Frobenius is a , recovering its parent uniquely.

If $a < b$ are effective generators of a parent, deleting a leaves b minimal, because deletion cannot introduce a decomposition. It also leaves b effective, because the child's Frobenius is a . Thus only deletion of the largest effective generator can give a leaf. At high depth, $F > m$; restoration and deletion of effective generators therefore preserve m . The coordinate statement follows from Lemma 2.1, and yields the last assertion. At multiplicity two a word (k) has effective generator $2k + 1$; in particular this boundary introduces no high-depth leaves. $\square$

All sets at a fixed genus are finite. Indeed the map $s \mapsto F - s$ injects the nongaps in $[0, F]$ into its gaps, whence $F \leq 2g - 1$ at positive genus g . Ordering the finite gap sets lexicographically, for example by their indicator vectors in $[1, 2g - 1]$, supplies an order whenever we use a finite-set matching below.

Fix $g \geq 1$. Partition the set $\mathcal{S}_g$ of genus- g semigroups by the following tests, applied in order; each test selects only sources not selected earlier:

- (1) $\mathcal{L}_g$: all low-depth words;
- (2) $\mathcal{P}_g$: all high-depth nonleaves;
- (3) $\mathcal{C}_g$: all remaining high-depth leaves with a low-depth parent;
- (4) $\mathcal{T}_g$: all remaining words $(5, u)$ with every letter of u at most three;
- (5) $\mathcal{D}_g$: all remaining words of depth at least six, together with all remaining depth-four or depth-five words containing no 1;

- (6) $\mathcal{I}_g$: all remaining words with a unique 5, at a position at least two, and all other letters at most three;
- (7) $\mathcal{B}_g$: all remaining depth-four words whose sole 1 is final;
- (8) $\mathcal{E}_g$: all remaining sources.

Lemma 2.3. *The exceptional set $\mathcal{E}_g$ is precisely the set of leaves containing at least one 1 in the following three disjoint sectors:*

- (1) *depth four with at least two 4's, excluding words whose sole 1 is final;*
- (2) *depth five with a unique 5 and at least one 4;*
- (3) *depth five with at least two 5's.*

Proof. The first two tests leave only high-depth leaves. By Lemma 2.2, the third removes exactly the depth-four leaves with a unique 4. The fifth removes all depths at least six and all words without 1. A depth-five word with only one letter at least four has a unique 5; it belongs to the fourth test if this is its first letter, and otherwise to the sixth. The seventh test makes the stated further exclusion at depth four. Every leaf in one of the three displayed sectors misses all seven tests, proving equality. $\square$

The append-1 and append-2 maps on low-depth words are the Eliahou–Fromentin injections [6, Propositions 5.2–5.3]. We recall their elementary proof because the change of multiplicity matters here. Appending either letter to a word of multiplicity m makes the new multiplicity $m + 1$. Old noncarry inequalities remain unchanged. The new noncarry target, at index m , is at most two; all its inequalities are automatic. Every new carry left side is at least three, and every target is at most three. Thus both append operations preserve validity.

Define the following maps:

$$\begin{aligned}
 (2.3) \quad & x \mapsto (x, 1), & x \in \mathcal{L}_g, \\
 & x \mapsto \text{the child obtained by deleting the largest effective generator}, & x \in \mathcal{P}_g, \\
 & x \mapsto (\text{par}(x), 2), & x \in \mathcal{C}_g, \\
 & (5, u) \mapsto (2, u, 1, 3), & (5, u) \in \mathcal{T}_g.
 \end{aligned}$$

Here $\text{par}(x)$ denotes the parent's Kunz word. For $x = (w, 1) \in \mathcal{B}_g$, replace each letter of w using

$$(2.4) \quad 2 \mapsto 11, \quad 3 \mapsto 21, \quad 4 \mapsto 22,$$

and append 2.

Lemma 2.4. *These maps are injective into $\mathcal{S}_{g+1}$ and their image sectors are pairwise disjoint. The $\mathcal{B}_g$ image consists of binary words ending in 2.*

Proof. The append-1 map is valid and reversible. A source in $\mathcal{C}_g$ has a depth-three parent of genus $g - 1$; appending 2 therefore gives a nonbinary word of genus $g + 1$. Removing this final letter determines the parent, and Lemma 2.2 determines its leaf child uniquely. The ordinary-child map is valid, has high depth, and is inverted by restoration of the Frobenius number.

For the fourth map, suppose the original multiplicity is m . The new multiplicity is $m + 2$, and the weight changes by $-5 + 2 + 1 + 3 = 1$. Old noncarry targets have value at most three. If an addend has changed, its new value is 2, so its sum with any positive letter is sufficient. The target at index m is 1. For the final target 3 at index $m + 1$, a pair of old indices had, in the source, a carry inequality targeting the initial 5, so its old sum was at least four. If one of these indices were 1, the other would be m , which is not an old index. Thus these sums are unchanged. The sole additional pair is $(1, m)$, of values $(2, 1)$. Every carry inequality is automatic at depth three. Removing the final 1, 3 and changing the initial 2 back to 5 gives the inverse.

In the last map, w is a word over $\{2, 3, 4\}$ with at least two 4's. Every block preserves weight, and replacement of the final 1 by 2 raises genus by one. Binary words satisfy all Kunz inequalities. Remove the final 2 and parse consecutive pairs to recover w uniquely.

The first four images are respectively low depth ending in 1, high depth, low depth and non-binary ending in 2, and low depth ending in 3. The code image is binary ending in 2. These descriptions are disjoint. $\square$

3. BINARY ALLOCATION

Write $F_0 = 0$, $F_1 = 1$, and $F_{g+1} = F_g + F_{g-1}$ for $g \geq 1$. There are exactly F_g binary words of weight $g + 1$ ending in 2: remove the final 2 and count compositions of $g - 1$ over $\{1, 2\}$. This includes $g = 1$, when the remaining composition is empty. We bound jointly the sources assigned to this sector, including its code images.

Define the following nonnegative formal series:

$$\begin{aligned} a_k(z) &= \sum_{[k/2] \leq a \leq k} z^a, & b_k(z) &= \sum_{\substack{1 \leq a, b \leq k \\ a+b \geq k}} z^{a+b}, & K_k(z) &= \frac{1 + a_k(z)}{1 - b_k(z)}, \\ a(z) &= \frac{z^4}{1 - z}, & b(z) &= \frac{z^8(7 - 6z)}{(1 - z)^2}. \end{aligned}$$

Put

$$(3.1) \quad M(z) = z^6 K_6 K_5 + z^7 K_7 K_6 + z^8 K_8 K_7 + \frac{z^9(1 + a)^2}{(1 - z)(1 - b)^2},$$

$$H(z) = \frac{z^4 + z^5}{(1 - z^2 - z^3)(1 - z^2 - z^3 - z^4 - z^5)},$$

$$J(z) = \frac{2z^{12} + 4z^{14} + 2z^{15}}{(1 - 2z^5 - z^6)(1 - z^2 - z^3)(1 - 3z^4 - 2z^5 - z^6)},$$

$$K(z) = \frac{z^9}{(1 - z^2 - z^3)^2(1 - z^2 - z^3 - z^4)},$$

$$(3.2) \quad W(z) = M(z) + H(z) + J(z) + K(z), \quad W_g = [z^g]W(z).$$

The notation $K(z)$ and $K_k(z)$ denotes different functions.

Lemma 3.1. *For every $g \geq 1$,*

$$|\mathcal{D}_g| + |\mathcal{I}_g| + |\mathcal{B}_g| \leq W_g.$$

Proof. At the rightmost maximum q , the coordinates to its left are paired by reflection about that position: each pair has sum at least q , and its letters are at most q . To its right, each pair has sum at least $q - 1$ and its letters are at most $q - 1$. A central self-pair has the corresponding ceiling-half lower bound. For either block, summing over even and odd lengths gives K_q or K_{q-1} . The two block lengths determine the pivot position, so $z^q K_q K_{q-1}$ bounds all words of depth q without an additional position factor. For $q \geq 9$, allow arbitrary positive letters, weaken both pair thresholds to eight and both central thresholds to four. The central series is a , and the ordered pair series is $\sum_{s \geq 8} (s - 1)z^s = b$. Summing the pivot weight over $q \geq 9$ yields the last term of (3.1). Thus M bounds depth at least six. Subtracting the composition series on $\{2, 3\}$ from that on $\{2, 3, 4, 5\}$ gives H , which counts all no-1 words of depth four or five. These two estimates bound $\mathcal{D}_g$.

For an isolated 5 at a noninitial position, the left block is nonempty, uses letters at most three, and has reflection-pair sum at least five. In particular it contains no 1. Its series is $L_0 - 1$, where

$$L_0 = \frac{1 + z^3}{1 - 2z^5 - z^6}.$$

The right block has pair threshold four and central threshold two, giving

$$Q_0 = \frac{1 + z^2 + z^3}{1 - 3z^4 - 2z^5 - z^6}.$$

Since the no-1 words were already removed, the right block must contain 1. Its subpopulation without 1 is exactly all $\{2, 3\}$ words, with series $C_0 = (1 - z^2 - z^3)^{-1}$. Direct simplification gives $z^5(L_0 - 1)(Q_0 - C_0) = J$, proving the required bound for $\mathcal{I}_g$.

Finally, splitting a $\{2, 3, 4\}$ word at its first two 4's gives the series

$$(3.3) \quad N(z) = \frac{z^8}{(1 - z^2 - z^3)^2(1 - z^2 - z^3 - z^4)}.$$

A final 1 supplies the factor z , so $K = zN$ majorizes $\mathcal{B}_g$. We discarded inequalities only to enlarge each population. Summing the three bounds proves the assertion. $\square$

Lemma 3.2 (Binary numerical bound). *For all $g \geq 0$, $W_g \leq 13F_g/25$. For $g \geq 141$, $W_g \leq 13F_g/1000$. Moreover, with $r = 809/500$,*

$$(3.4) \quad \frac{W_g}{F_g} < \frac{50r^2}{((63/100)r)^g} \quad (g \geq 1).$$

Proof. The exact finite arithmetic inputs are

$$\begin{aligned} 25W_g &\leq 13F_g & (0 \leq g \leq 499), \\ 1000W_g &\leq 13F_g & (141 \leq g \leq 499), \\ W(63/100) &< 50, & \frac{50r^2}{((63/100)r)^{500}} < \frac{13}{1000}. \end{aligned}$$

All geometric denominators in (3.2) are positive at $63/100$. These are finite rational checks. In particular, for each rational summand $P(z)/Q(z)$, with $Q(0) = 1$, its coefficients are calculated using

$$(3.5) \quad f_n = [z^n]P - \sum_{j=1}^{\min(n, \deg Q)} [z^j]Q f_{n-j}.$$

This recurrence and the displayed rational formulas specify the exact arithmetic checks without numerical rounding. Since $r^2 < r + 1$, the two initial cases and induction in the Fibonacci recurrence give $F_g \geq r^{g-2}$ for $g \geq 1$. Nonnegativity gives $W_g(63/100)^g \leq W(63/100) < 50$, proving (3.4). The base $(63/100)r$ is greater than one, so the sharper finite bound extends to every $g \geq 500$ and also implies the weaker bound there. $\square$

Reserve the actual images (2.4) first. In the fixed finite-set orders, match $\mathcal{D}_g \cup \mathcal{I}_g$ to the binary end-2 words not reserved. Lemmas 3.1 and 3.2 show that the total demand, including reserved images, is at most $13F_g/25$, hence at most the supply F_g . This matching is injective and its images are disjoint from all images in Lemma 2.4. We have proved the following.

Proposition 3.3. *There is an injection $\Phi_g : \mathcal{S}_g \setminus \mathcal{E}_g \longrightarrow \mathcal{S}_{g+1}$. At least*

$$(3.6) \quad L_{\text{bin}}(g) = \max \left\{ \left\lceil \frac{12F_g}{25} \right\rceil, F_g - W_g \right\}$$

binary targets of genus $g + 1$ ending in 2 lie outside its image.

The two entries in this maximum bound the same unused set. In particular, they must not be added. For $g \geq 141$, this gives $L_{\text{bin}}(g) \geq 987F_g/1000$.

4. WITNESSED UNUSED TARGETS

Let a prefix P of length $n \geq 1$, ending in 3, be followed by a binary tail of length ℓ . The base word has multiplicity $m = n + \ell + 1$. A prefix 3 at position $i = n - d$ is *witnessed* if two tail positions $n + j, n + k$, both containing 1, satisfy

$$(4.1) \quad j + k = \ell + 1 - d.$$

We permit $j = k$. This is exactly the carry equality $(n + j) + (n + k) = m + i$. Hence replacing that 3 by 4 violates its carry inequality, whose left side is $1 + 1 + 1 = 3$.

Lemma 4.1. *If a base word is valid and every 3 is witnessed, appending 2 gives a nonbinary target outside the image of Φ_g , where g is one greater than the weight of the base. Distinct bases give distinct unused targets.*

Proof. The base has depth three, so its Frobenius exceeds its multiplicity. Its ordinary children preserve multiplicity, and a high-depth child would have to increase a coordinate 3 to 4. Every such increase is blocked by (4.1), so no such child exists. Appending 2 preserves validity. A target in the $\mathcal{C}_g$ image would recover, on removing this 2, a base having a high-depth leaf child, a contradiction. Depth, final letter, and the presence of 3 exclude every other image sector. Deleting the appended 2 recovers the base. Its weight is $g - 1$, while the target weight is $g + 1$, giving the stated genus shift. $\square$

Put $u = z^2 + z^3$, $s = z + z^2 + z^3$, $c = z + z^2$, and $v = us$. We use two disjoint families of prefixes. A *safe-half prefix* of length n has its first $\lfloor n/2 \rfloor$ letters in $\{2, 3\}$, its remaining nonfinal letters in $\{1, 2, 3\}$, and its last letter 3. This is Bacher's family in the proof of [1, Theorem 15.1, Section 15.1]. We use it here with tail witnesses to obtain targets outside the partial injection; the prefix family itself is not new. An *early-single-one prefix* has its unique prefix 1 at a position $p \geq 1$, has 2 at position $2p$, ends in 3 at some $n \geq 2p + 1$, and has all other letters in $\{2, 3\}$.

Every base obtained from either family and an arbitrary binary tail is valid. At depth three only a noncarry $1 + 1$ targeting 3 can fail. In the safe-half family two prefix 1 positions sum beyond the prefix. In the early-single-one family the only prefix pair of 1's is the self-pair at p , targeting the prescribed 2 at $2p$. A pair involving a tail position cannot noncarry-target the prefix. These observations prove validity. The early 1 lies in the first half, so the two families are disjoint. The last 3 uniquely identifies the prefix in a base.

Writing $U = \{2, 3\}$ and $V = \{1, 2, 3\}$, the safe prefixes have forms $U^k V^k 3$ and $U^{k+1} V^k 3$, where powers denote independently chosen letters. The early prefixes have form $U^h 1 U^h 2 U^k 3$; the two U^h blocks need not be identical. Therefore their weight series are

$$(4.2) \quad C_s = \frac{z^3(1+u)}{1-v}, \quad C_e = \frac{z^6}{(1-u)(1-u^2)}.$$

We now count bases with at least one unwitnessed 3 by marking a failed 3. Let t record the distance of the mark from the prefix end. The two marked-prefix series are

$$(4.3) \quad F_s(z, t) = C_s + \frac{z^6 u(1+u)t}{(1-v)(1-vt)} + \frac{z^6 t(1+st)}{(1-vt)(1-vt^2)},$$

$$(4.4) \quad F_e(z, t) = C_e + \frac{z^9 t}{(1-u^2)(1-u)(1-ut)} + \frac{z^9 ut^2}{(1-ut)(1-u^2)(1-u^2t)} + \frac{z^9 ut^4}{(1-ut)(1-u^2t)(1-u^2t^2)}.$$

To verify these formulas directly, marking the terminal 3 contributes C_s or C_e . In a block of length k with letter-weight a , marking a 3 with j letters to its right replaces its block weight by $z^3 a^{k-1} t^j$, for $0 \leq j < k$. Multiply by t to the total length of subsequent blocks. In the safe forms

above, marking the V^k blocks gives the second term of (4.3); marking the first U blocks gives the third. In the early form, mark in order the U^k block, the second U^h block, and the first U^h block. These give the three nonterminal terms of (4.4), respectively. Summing each block by the finite geometric identity $\sum_{j=0}^{k-1} t^j = (1 - t^k)/(1 - t)$ yields the displayed expressions. Thus these nonnegative series mark actual occurrences, with multiplicity when a prefix has several 3's.

For a mark at distance d , let $Q_d(z)$ count binary tails that fail to witness it. Then

$$(4.5) \quad Q_d(z) = \sum_{0 \leq \ell < d} c^\ell + c^d A_w(z), \quad A_w(z) = \frac{1 + z^2}{1 - 2z^3 - z^4}.$$

Indeed no witness is possible for $\ell < d$. For $\ell \geq d$, the first $\ell - d$ positions are paired by reflection. A pair cannot be 11, so its possibilities are 12, 21, 22, of weight $2z^3 + z^4$; a central singleton must be 2. The last d positions are arbitrary. The empty constrained block when $\ell = d$ is included by the numerator of A_w .

For a marked series $F(z, t) = \sum_{d \geq 0} F_d(z) t^d$, define

$$(4.6) \quad \mathcal{U}(F) = z \left\{ \frac{F(z, 1) - F(z, c)}{1 - c} + A_w F(z, c) \right\} = z \sum_{d \geq 0} F_d(z) Q_d(z).$$

This is a nonnegative formal series counting failed marks. It upper bounds bad bases, since each bad base has at least one failed mark. The extra factor z converts base weight into the source genus in Lemma 4.1. Set

$$(4.7) \quad T_s = \frac{zC_s}{1 - c}, \quad T_e = \frac{zC_e}{1 - c}, \quad U_s = \mathcal{U}(F_s), \quad U_e = \mathcal{U}(F_e).$$

$$L_{\text{nonbin}}(g) = \max\{0, [z^g](T_s - U_s)\} + \max\{0, [z^g](T_e - U_e)\}.$$

Separate clipping is legitimate because the two base families are disjoint. Lemma 4.1 proves that this is a lower bound on the number of unused nonbinary targets.

Lemma 4.2. *For $g \geq 305$, $L_{\text{nonbin}}(g) \geq (927/500)F_g$.*

Proof. Define nonnegative error series

$$E_s = \frac{(1 + z)(1 + z^2)^2}{1 - v}, \quad E_e = \frac{z^4(1 + z + z^2)}{(1 - u)(1 - u^2)}, \quad S = E_s + E_e + U_s + U_e.$$

Direct multiplication of denominators gives

$$T_s = \frac{1}{1 - c} - E_s,$$

$$T_e = \frac{z^4}{1 - c} - E_e + \frac{z^9(2 + 2z + z^2)}{(1 - u)(1 - u^2)}.$$

As $[z^j](1 - z - z^2)^{-1} = F_{j+1}$, the sum before clipping is at least $F_{g+1} + F_{g-3} - [z^g]S$ for $g \geq 4$. All geometric denominators in $E_s, E_e, A_w, F_s(z, c), F_e(z, c)$ are positive at $z = 5/8$. Although $c(5/8) > 1$, the expression for $\mathcal{U}$ is convergent there: for this positive value of c , $\sum_{\ell < d} c^\ell \leq c^d/(c - 1)$, so (4.6) is dominated by a constant times $F(z, c)$. We are not expanding $(1 - c)^{-1}$ as a convergent geometric series at this point.

The exact rational inputs, computed by (3.5), are

$$S(5/8) < 78,$$

$$78(809/500)^2(800/809)^{1500} < 1/10000,$$

$$10000(F_{g+1} + F_{g-3}) \geq 18541F_g \quad (g = 16, 17),$$

$$500L_{\text{nonbin}}(g) \geq 927F_g \quad (305 \leq g \leq 1499).$$

The difference in the third line satisfies the Fibonacci recurrence, so the inequality holds for every $g \geq 16$. Nonnegative coefficients and $F_g \geq (809/500)^{g-2}$ give

$$\frac{[z^g]S}{F_g} < 78(809/500)^2(800/809)^g < \frac{1}{10000} \quad (g \geq 1500).$$

Subtracting this error gives $18540F_g/10000 = 927F_g/500$ for $g \geq 1500$. The finite interval in the last line completes the proof. $\square$

Proposition 4.3. *Define the exact integer lower bound*

$$(4.8) \quad L(g) = L_{\text{bin}}(g) + L_{\text{nonbin}}(g).$$

At least $L(g)$ low-depth targets of genus $g+1$ are outside the image of Φ_g . For $g \geq 305$,

$$(4.9) \quad L(g) \geq \frac{2841}{1000}F_g,$$

$$(4.10) \quad \frac{L(g)}{F_g} \geq 1 - \frac{50(809/500)^2}{((63/100)(809/500))^g} + \frac{927}{500}.$$

Proof. The binary and nonbinary unused sets are disjoint. Thus their bounds add. The first estimate combines $987/1000$ and $927/500$; the second combines $L_{\text{bin}}(g) \geq F_g - W_g$ with (3.4) and Lemma 4.2. $\square$

5. THE REMAINING COUNTING PROBLEM

Let t_g count low-depth genus- g words, and write $c_g = |\mathcal{C}_g|$, $f_g = |\mathcal{T}_g|$, $d_g = |\mathcal{D}_g|$, $i_g = |\mathcal{I}_g|$, $b_g = |\mathcal{B}_g|$. Only the ordinary-child map uses high-depth targets. The exact number of unused low-depth targets is therefore

$$(5.1) \quad R_g = t_{g+1} - t_g - c_g - f_g - d_g - i_g - b_g \geq L(g).$$

Counting the $n_g - |\mathcal{E}_g|$ matched targets and these unused targets proves the fundamental inequality

$$(5.2) \quad n_{g+1} - n_g \geq R_g - |\mathcal{E}_g| \geq L(g) - |\mathcal{E}_g|.$$

There may also be unused high-depth targets, which only strengthen it.

Let A_g count all genuine depth-four words with at least two 4's, and let B_g count all genuine depth-five words with at least two 5's. No leaf or 1 condition is imposed in these two definitions.

Lemma 5.1. *The unique-5 sector of $\mathcal{E}_g$ injects into nonleaf depth-four words of genus $g-1$ with at least two 4's and at least one 1. Consequently $|\mathcal{E}_g| \leq A_g + A_{g-1} + B_g$.*

Proof. Restore the Frobenius, changing the unique 5 to 4. The result is its ordinary parent, has one new and at least one preexisting 4, and retains its 1's. It is a nonleaf because the source is its child. Two such sources cannot have the same parent by Lemma 2.2. The other two sectors of Lemma 2.3 are directly counted by A_g and B_g . $\square$

For $q \in \{4, 5\}$ define the *raw cap- q population* as follows. Choose a word with at least two q 's and denote their two rightmost positions by $r < p$. Coordinates before r belong to $\{1, \dots, q\}$; all nonpivot coordinates after r belong to $\{1, \dots, q-1\}$. Retain only the Kunz inequalities targeting r or p , including all self-pairs. Every genuine word counted by A_g or B_g belongs to the corresponding raw population. The two rightmost maxima are unique, so summing over their positions introduces no multiple counting.

Proposition 5.2. *Let U_g be any integer upper bound on the complete raw cap-four coefficient of weight g , and let V_g be any integer upper bound on B_g . With $N_g = [z^g]N(z)$ from (3.3), the quantity*

$$(5.3) \quad E_{\text{upper}}(g) = U_g - N_g - N_{g-3} - N_{g-4} + U_{g-1} - N_{g-1} + V_g$$

is an upper bound on $|\mathcal{E}_g|$.

Proof. Raw cap-four words without 1 are exactly the $\{2, 3, 4\}$ words with at least two 4's, and every such word is genuine: noncarry sums are at least four and carry sums at least five. Their exact series is $N(z)$. For a word whose sole 1 is final, all inequalities are automatic except its self-pair, which is a carry pair targeting the penultimate coordinate and forces that coordinate to be at most three. If the penultimate coordinate were 4, it would be the rightmost maximum, so the raw population retains precisely this obstruction. Thus raw and genuine populations of this description coincide and consist of $(v, 2, 1)$ and $(v, 3, 1)$ with v a $\{2, 3, 4\}$ word containing at least two 4's. Their series is $(z^3 + z^4)N(z)$, disjoint from the no-1 population. Subtracting these exact counts from U_g bounds the first residual sector.

Lemma 5.1 bounds the second sector by the raw cap-four count of weight $g - 1$ containing 1, hence by $U_{g-1} - N_{g-1}$. Its image need not be a residual leaf and may have its sole 1 final, so there is no further subtraction from this shifted term. Finally V_g bounds the third sector. Adding these bounds proves the claim. The exact exclusions are from the whole raw population; the way an upper bound U_g is assembled does not affect this subtraction. $\square$

For example, $(5, 4, 2, 1)$ is a leaf with parent $(4, 4, 2, 1)$, whose sole 1 is final. The latter has effective generator 21, and its deletion gives the former. Thus the absence of a final-1 exclusion in the shifted term of (5.3) is essential.

We will also use the following real-variable improvement. Let $D(z)$ count the first two residual sectors together, with their actual genus weights, and put $A(z) = \sum_g A_g z^g$, $B(z) = \sum_g B_g z^g$.

Lemma 5.3. *At every $0 < z \leq 1$ for which $A(z)$ converges, $D(z) \leq A(z)$. In particular, whenever $0 < z_4, z_5 \leq 1$ and both masses are finite,*

$$(5.4) \quad |\mathcal{E}_g| \leq A(z_4)z_4^{-g} + B(z_5)z_5^{-g}.$$

Proof. Map each depth-four residual leaf to itself, and each unique-5 residual leaf to its parent as in Lemma 5.1. The former images are leaves and the latter are nonleaves, so they are disjoint. This gives an injection into the population counted by A . The first map preserves genus, while the second lowers it by one. For $0 < z \leq 1$ each source weight is therefore at most its image weight. Nonnegative summation proves $D(z) \leq A(z)$. Extract a single nonnegative coefficient at z_4 , and bound the remaining multiple-5 sector using B at z_5 , to obtain the result. This is a positive-real comparison, not a coefficientwise assertion $[z^g]D \leq A_g$. $\square$

6. A COEFFICIENT BOUND FOR THE RAW TWO-PIVOT TAIL

This section concerns a relaxation of the Kunz inequalities. Its objects need not represent numerical semigroups. Transfer matrices and rational generating series are established tools in the enumeration of numerical semigroups; see Bacher [1]. The point here is a uniform coefficient estimate after summing over both the multiplicity and an unbounded set of pivot positions.

Definition 6.1 (Raw two-pivot polynomial). For integers $1 \leq r < p < m$ and $a \in \{4, 5\}$, let $Z_{m,r,p}^{(a)}(z)$ count words $(x_1, \dots, x_{m-1})$ with

$$x_r = x_p = a, \quad 1 \leq x_i \leq a \ (i < r), \quad 1 \leq x_i \leq a - 1 \ (i > r, i \neq p),$$

weighted by $z^{\sum_i x_i}$. Retain exactly the inequalities

$$x_i + x_j \geq a \ (i + j \in \{r, p\}), \quad x_i + x_j \geq a - 1 \ (i + j \in \{m + r, m + p\}),$$

including the cases $i = j$. No inequality with another target is imposed. Write $Z_{m,r,p} = Z_{m,r,p}^{(4)}$ and

$$F_{r,p}(z) = \sum_{m > p} Z_{m,r,p}(z).$$

The pivots are uniquely the two rightmost occurrences of the maximum. Consequently summing over m, r, p counts each raw word once. A genuine depth- a Kunz word with at least two maxima belongs to this relaxation.

Theorem 6.2 (Raw omitted-pivot estimate). *For every integer $g \geq 0$,*

$$(6.1) \quad \sum_{p > 300} \sum_{r=1}^{p-1} [z^g] F_{r,p}(z) \leq \frac{221715}{1000000} F_g.$$

The left side vanishes for $g < 604$. No leaf condition or occurrence of a coordinate equal to one is required.

6.1. The reflection graph and its support. Remove the two fixed pivots. A noncarry pair gives a *strong* edge, of threshold a , and a carry pair gives a *carry* edge, of threshold $a - 1$. A self-pair is a unary restriction, represented as a loop rather than an ordinary edge. Each vertex has at most one reflection incidence for each target. Thus the ordinary graph has maximum degree two and its components are paths, isolated vertices, and cycles. A cycle alternates the two targets modulo m , has even length, and has length at least four: distinct targets cannot give parallel edges. Constraints incident to a fixed pivot are automatic.

This description also specifies the exact polynomial: sum the products of vertex weights over allowed assignments along each path; for a cycle, fix one starting value and close the last edge back to that value. Multiply the component polynomials and the pivot factor z^{2a} . Loops impose their unary restrictions at the corresponding endpoints. An isolated vertex is a path of zero ordinary-edge length.

Lemma 6.3 (Finite support). *Every word counted by $Z_{m,r,p}^{(a)}$ has genus satisfying*

$$(6.2) \quad 2g \geq (a - 1)m + p + 1.$$

In particular, for cap four, $g \geq 2p + 2$ and $g \geq m + 5$.

Proof. The reflection targeting p pairs the $p - 1$ coordinates to its left with sum at least a and the $m - p - 1$ coordinates to its right with sum at least $a - 1$. A central fixed point satisfies twice its coordinate at least the same threshold. Adding these inequalities, with twice the pivot weight, gives

$$2g \geq a(p - 1) + (a - 1)(m - p - 1) + 2a = (a - 1)m + p + 1.$$

For $a = 4$ use $m \geq p + 1$. Independently, the two pivot entries contribute eight and the other $m - 3$ coordinates contribute at least one each, giving $g \geq m + 5$. $\square$

All series and coefficient sums below are therefore well defined formally. The support assertion in Theorem 6.2 follows already from $p \geq 301$.

6.2. A structural recurrence and cancellation of pure modes. Fix $r < p$, and put $d = p - r$. For $m > 2p$, temporarily include the pivot p as an ordinary vertex of the carry graph. Set $x = i - r$ and $N = m - r$. The carry vertices are $1, \dots, N - 1$, and their two reflections are

$$(6.3) \quad \mathcal{F}(x) = N - x, \quad \mathcal{G}(x) = N + d - x.$$

The first is defined throughout this interval. The second stays in the interval exactly when $x > d$. Thus the vertices $1, \dots, d$ have one carry incidence each and form the *ports*. The pivot p is port d ; the surviving ports correspond to coordinates $r + 1, \dots, p - 1$. Their alphabets are all $\{1, 2, 3\}$.

Here is the complete path decomposition, including the loops. For a representative $a \in \{1, \dots, d\}$, list its residue class in the interval as $a_k = a + kd$, $0 \leq k < n_a$. Let $b \in \{1, \dots, d\}$

be determined by $a + b \equiv N \pmod{d}$. The map $\mathcal{F}$ reverses the two lists and bijects them, so $n_a = n_b = n$. Since $a + (n-1)d + b = N$, their index formulas are

$$\mathcal{F}(a_k) = b_{n-1-k}, \quad \mathcal{G}(a_k) = b_{n-k} \quad (1 \leq k < n).$$

For $a \neq b$, alternating these formulas gives the path

$$a_0, b_{n-1}, a_1, b_{n-2}, \dots, a_{n-1}, b_0.$$

It contains all $2n$ vertices in the two classes, has ports a, b , and has $2n-1$ ordinary edges. For $a = b$, the same alternation visits the indices $0, n-1, 1, n-2, \dots$ until all n vertices have appeared. It has one port, $n-1$ ordinary edges, and a loop at its other endpoint. That loop belongs to $\mathcal{F}$ when n is odd and to $\mathcal{G}$ when n is even. Both loops require the coordinate value to be at least two. The orbits of $a \mapsto N - a \pmod{d}$ exhaust the residue classes, so this list omits no carry component.

Replacing m by $m+d$ leaves the port involution unchanged and adds one vertex to every residue class. Each paired-class path gains two ordinary edges, and each fixed-class path gains one. If h_c is the number of paired-class paths and f the number of fixed classes, then

$$2h_c + f = d,$$

so the total carry-edge increment is exactly d . A change of loop target does not change its unary restriction. Deleting port d removes an endpoint and its incident ordinary edge, never an internal vertex. It creates one free endpoint. The strong graph, its loops and the alphabets on all surviving ports are independent of m .

The symmetric cap-three carry matrix is

$$C(z)_{ij} = z^{(i+j)/2} \mathbf{1}_{i+j \geq 3} \quad (1 \leq i, j \leq 3).$$

It has rank two. Its nonzero eigenvalues λ, μ satisfy

$$(6.4) \quad \lambda + \mu = b = z^2 + z^3, \quad \lambda\mu = c = -z^3 - z^4.$$

For a positive ordinary-edge length ℓ , its path kernel is $C^\ell = \lambda^\ell P_\lambda + \mu^\ell P_\mu$. There is one possible zero-edge case to check. Since $m > 2p$, every residue class has at least two vertices. Consequently a paired-class path still has positive length after deleting p , whereas a fixed-class path with two vertices becomes one free, loop-constrained vertex. The free and loop boundary vectors for this component are

$$h = (z^{1/2}, z, z^{3/2})^T, \quad h_+ = (0, z, z^{3/2})^T.$$

They both belong to the image of C , which is the span of $(1, 0, 0)^T$ and $(0, z, z^{3/2})^T$. Therefore

$$h^T(P_\lambda + P_\mu)h_+ = h^T h_+.$$

The zero-edge component is thus also exactly the sum of its two nonzero modes. It introduces no extra transient term into the recurrence.

Lemma 6.4 (Fixed-pivot rationality). *On each progression $m = m_0 + kd$ with $m_0 > 2p$, the polynomials $Z_{m,r,p}$ satisfy the scalar recurrence with characteristic polynomial*

$$P_d(T, z) = \prod_{j=0}^d (T - \lambda^{d-j} \mu^j) \in \mathbb{Z}[z, T].$$

Consequently $F_{r,p}$ is a rational function.

Proof. On this progression the port pairing, endpoint restrictions and strong core are fixed. Expand each carry kernel into its two spectral summands and contract against that core. A paired-class component assigned the λ mode contributes a factor λ^{2k} under the shift $m_0 \mapsto m_0 + kd$; a fixed-class component contributes λ^k . The same statements hold with μ in place of λ . If j is

the sum of the increments assigned to μ , the remaining increments sum to $d - j$. Collecting the finitely many mode assignments therefore gives

$$Z_{m_0+kd,r,p} = \sum_{j=0}^d A_j(z) (\lambda^{d-j} \mu^j)^k.$$

This identity holds over the corresponding algebraic function field, including at $k = 0$ by the zero-edge calculation above; its amplitudes need not be positive. Applying $\prod_{j=0}^d (T - \lambda^{d-j} \mu^j)$ to this exponential sum proves the recurrence. For integrality of the annihilator, put $S_0 = 2$, $S_1 = b$, and $S_n = bS_{n-1} - cS_{n-2}$. Pair the roots indexed by $j, d - j$ to obtain

$$T^2 - c^j S_{d-2j} T + c^d \quad (0 \leq j < d/2),$$

with the additional factor $T - c^{d/2}$ when d is even.

For an explicit rational construction define

$$Q_d(y, z) = \prod_{j=0}^d (1 - \lambda^{d-j} \mu^j y) = \sum_{j=0}^{d+1} q_j(z) y^j.$$

For $1 \leq s \leq d$ set $m_s = 2p + s$ and

$$N_s(y, z) = \sum_{k=0}^d \sum_{j=0}^k q_j(z) Z_{m_s+(k-j)d,r,p}(z) y^k.$$

The recurrence yields

$$(6.5) \quad F_{r,p}(z) = \sum_{m=p+1}^{2p} Z_{m,r,p}(z) + \frac{\sum_{s=1}^d N_s(1, z)}{Q_d(1, z)}.$$

The substitution $y = 1$ is justified coefficientwise by $g \geq m + 5$, and $Q_d(1, 0) = 1$. In particular this is a structural recurrence, not a recurrence fitted to finitely many values. $\square$

Proposition 6.5 (Cancellation and numerator degree). *Let $\tau = 1$ when d is odd and $\tau = 2$ when d is even. Then*

$$(6.6) \quad H_{r,p} := (1 - z - z^2) F_{r,p} = \frac{A_{r,p}}{B_d}, \quad B_d = K_\tau R_d,$$

where $A_{r,p} \in \mathbb{Q}[z]$, $\deg A_{r,p} \leq 3d^2 + 7p + 6$, and

$$\begin{aligned} K_1 &= 1 + z + z^2, \\ K_2 &= 1 + z + 2z^2 + z^3 - z^5 - z^6, \\ R_d &= \prod_{j=1}^{d-1} (1 - \lambda^{d-j} \mu^j) \in \mathbb{Z}[z]. \end{aligned}$$

Proof. We first separate the multiplicity dependence of a pure mode from the way in which its ports are paired. Fix $\theta \in \{\lambda, \mu\}$, choose a nonzero eigenvector v_θ of the symmetric carry matrix, and put $t_\theta = v_\theta^T v_\theta$. Over an algebraic function field these choices can be made with $t_\theta \neq 0$, and

$$P_\theta = \frac{v_\theta v_\theta^T}{t_\theta}.$$

Let $\mathcal{C}_{r,p}(v_\theta)$ be the contraction of the fixed strong core, including its strong loops and the factor z^8 for the pivots. At each surviving port with coordinate value i , use the weight $z^{i/2}(v_\theta)_i$; use the full weight z^i at each other core vertex. The other half of each port weight is already in the adjacent symmetric carry kernel. This definition specifies one fixed scalar, independent of m and of the port pairing.

There are $d - 1$ surviving ports, one free carry endpoint created by deleting p , and f carry-loop endpoints. There are $(d + f)/2$ carry components, each contributing one rank-one projector. Factoring its two endpoint entries shows that the entire pure- θ contribution is exactly

$$(6.7) \quad \theta^{e_C(m)} \mathcal{C}_{r,p}(v_\theta) \frac{(h^T v_\theta)(h_+^T v_\theta)^f}{t_\theta^{(d+f)/2}}.$$

For example, a paired-port kernel contributes $\theta^\ell(v_\theta)_i(v_\theta)_j/t_\theta$; a port-to-loop kernel replaces one endpoint entry by $h_+^T v_\theta$. The component containing the deleted port instead has the factor $h^T v_\theta$ at its free endpoint. The zero-edge free-to-loop component has these same factors, by its full two-projector decomposition in Section 6.2. Thus every surviving port contributes exactly one entry of v_θ , regardless of its partner. Formula (6.7) is unchanged if v_θ is rescaled, and proves the asserted independence from pairing.

For $m > 2p$, counting the two carry reflections and deleting the ordinary edge at p gives

$$e_C(m) = \left\lfloor \frac{m - r - 1}{2} \right\rfloor + \left\lfloor \frac{m - p - 1}{2} \right\rfloor - 1.$$

The fixed classes solve $2a \equiv m - r \pmod{d}$. When d is odd, there is one such class, so $f = 1$ and $e_C = m - (r + p + 5)/2$. All factors in (6.7) other than $\theta^{e_C(m)}$ are therefore constant as m increases by one. Summing the pure terms has denominator $1 - \theta$. When d is even, the values are

	f	$e_C(m)$
$m - r$ odd	0	$m - (r + p)/2 - 2$
$m - r$ even	2	$m - (r + p)/2 - 3$

and splitting the sum by parity gives denominator $1 - \theta^2$. In either case the two pure contributions together have a denominator dividing E_τ , where

$$E_j = (1 - \lambda^j)(1 - \mu^j) = 1 - S_j + c^j, \quad L_d = E_d/E_\tau, \quad \Delta = b^2 - 4c.$$

Since $\tau \mid d$, the quotient L_d is a polynomial: it is the product of the two finite geometric sums for λ^τ and μ^τ , which is symmetric in λ, μ .

We next descend this mode calculation to rational functions with controlled denominators. This avoids assuming that the roots being cancelled are simple. The rational-weight carry matrix is

$$M = \begin{pmatrix} 0 & z^2 & z^3 \\ z & z^2 & z^3 \\ z & z^2 & z^3 \end{pmatrix}.$$

It is similar to C and assigns the full vertex weight to the arriving vertex of an oriented edge. Its nonzero spectral projector is

$$\Pi_\lambda = \frac{M(M - \mu I)}{\lambda(\lambda - \mu)},$$

with the analogous formula for Π_μ . Set

$$\mathcal{A} = \mathbb{Q}[z, (c\Delta)^{-1}], \quad \mathcal{S} = \mathcal{A}[\lambda].$$

The identities $\lambda^{-1} = \mu/c$ and $(\lambda - \mu)^{-1} = (\lambda - \mu)/\Delta$ put both projectors in $\text{Mat}_3(\mathcal{S})$. All strong-core and boundary weights in this rational convention also belong to $\mathcal{S}$; any redistribution of a port weight uses only powers of z , which are invertible in $\mathcal{A}$. Thus finite mode amplitudes have no possible denominators outside this localization.

The polynomial $\Delta = z^3(z + 1)(z^2 + z + 4)$ is not a square in $\mathbb{Q}(z)$, so

$$\mathcal{S} = \mathcal{A} \oplus \mathcal{A}\lambda.$$

The conjugation $\sigma(\lambda) = \mu = b - \lambda$ has fixed ring $\mathcal{S}^\sigma = \mathcal{A}$: if $a_0 + a_1\lambda$ is fixed, then $a_1(\lambda - \mu) = 0$, hence $a_1 = 0$. For the pure terms, the preceding summation has denominator dividing E_τ . For

a mixed assignment on any multiplicity progression, Lemma 6.4 gives a geometric denominator $1 - \lambda^{d-j}\mu^j$ with $1 \leq j \leq d-1$, which is a factor of R_d . It follows that $E_\tau R_d F_{r,p}$ belongs to $\mathcal{S}$. The automorphism σ of $\mathbb{Q}(z)(\lambda)$ fixes $\mathbb{Q}(z)$ pointwise. By Lemma 6.4, $F_{r,p} \in \mathbb{Q}(z)$, while $E_\tau, R_d \in \mathbb{Q}[z]$; hence their product is fixed by σ . Therefore

$$(6.8) \quad E_\tau R_d F_{r,p} \in \mathcal{A}.$$

On the other hand, (6.5) writes $F_{r,p} = N/(E_d R_d)$ with $N \in \mathbb{Q}[z]$. Equation (6.8) says precisely that N/L_d belongs to $\mathcal{A}$. To conclude polynomial divisibility, we check $\gcd(L_d, c\Delta) = 1$. At $z = 0, -1$ both eigenvalues vanish and $E_d = 1$. At the other discriminant zeros, $z^2 + z + 4 = 0$, the eigenvalues coincide at $b/2 = -2z$ and have modulus four, so again $E_d \neq 0$. Since L_d divides E_d , it has no zero at any zero of $c\Delta$. Write $N/L_d = G/(c\Delta)^k$ in (6.8). Then $N(c\Delta)^k = GL_d$, and coprimality implies $L_d \mid N$ in $\mathbb{Q}[z]$, with every root multiplicity. This argument does not require L_d and R_d to be coprime. Finally

$$E_\tau = (1 - z - z^2)K_\tau$$

gives (6.6) with $A_{r,p} = N/L_d$.

For the degree bound, the maximum allowed weight of a graph word is

$$8 + 4(r-1) + 3(m-r-2) = 3m + r - 2.$$

In the paired-factor formula for $Q_d(y, z)$, the coefficients of y and y^2 have degrees at most $3d$ and $4d \leq 6d$, respectively; the optional middle factor has y -coefficient of degree $2d \leq 3d$. Consequently $\deg q_j \leq 3dj$. A summand in N_s has degree at most

$$3dj + 3(m_s + (k-j)d) + r - 2 \leq D_0 := 3d^2 + 3d + 6p + r - 2,$$

since $m_s \leq 2p + d$ and $k \leq d$. Each paired factor of $Q_d(1, z)$ has degree $4d$, and the optional middle factor has degree $2d$, giving $\deg Q_d(1, z) = 2d(d+1)$. Thus the finite prefix in (6.5), after multiplication by this denominator, contributes degree at most

$$2d(d+1) + 6p + r - 2 \leq D_0.$$

Finally $\deg E_j = 4j$, so cancelling L_d removes degree $4(d-\tau)$. Substituting $r = p - d$ yields

$$\deg A_{r,p} \leq 3d^2 - 2d + 7p + 4\tau - 2 \leq 3d^2 + 7p + 6.$$

□

6.3. Uniform absolute coefficient norms. Set $q = (\sqrt{5} - 1)/2$, and for a formal series f write $\|f\|_q = \sum_{j \geq 0} |[z^j]f|q^j$, allowing the value infinity. This norm is submultiplicative whenever the right side is finite.

Lemma 6.6 (Uniform denominator bounds). *For every $d \geq 1$,*

$$(6.9) \quad \|B_d\|_q < 10, \quad \|B_d^{-1}\|_q < 21.$$

Proof. At q the nonzero carry eigenvalues are $1, -q^2$. Put $u = q^2$. The polynomials S_n have nonnegative coefficients, since $b, -c$ do. For a paired mixed factor, $j \geq 1$ and $n = d - 2j \geq 1$, its nonconstant coefficient norm is at most

$$u^j S_n(q) + u^d \leq u^j + u^{d-j} + u^d \leq a_j := u^j + u^{j+1} + u^{2j+1}.$$

Let

$$a_1 = u + u^2 + u^3, \quad T = \sum_{j \geq 2} a_j = \frac{(1+u)u^2}{1-u} + \frac{u^5}{1-u^2}.$$

Both a_1, T are less than one. An optional central factor $1 - c^{d/2}$ costs at most $1 + u$ in norm and $(1 - u)^{-1}$ in inverse norm. Geometric-series bounds and the finite-product inequality $\prod(1 - a_j) \geq 1 - \sum a_j$ give

$$(6.10) \quad \|R_d\|_q \leq \frac{(1+u)(1+a_1)}{1-T} < \frac{10}{3}, \quad \|R_d^{-1}\|_q \leq \frac{1}{(1-u)(1-a_1)(1-T)} < 6.$$

These comparisons follow on replacing q by the larger rational number $q_+ = 309017/500000$, since the displayed upper bounds increase with u .

For completeness the fixed-factor reciprocal estimates can be certified by a short, precisely specified finite calculation. For either coefficient list

$$(k_0, \dots, k_e) = (1, 1, 1) \quad \text{or} \quad (1, 1, 2, 1, 0, -1, -1),$$

put $v_0 = 1$, $v_n = -\sum_{j=1}^{\min(e,n)} k_j v_{n-j}$ for $1 \leq n \leq 80$, $V(z) = \sum_{n=0}^{80} v_n z^n$, and $E = 1 - KV$. The deliberately rounded rational inequalities in Table 2 are sufficient:

TABLE 2. Rational bounds for the degree-80 reciprocal polynomials and their residuals at $q_+ = 309017/500000$.

	$\ V\ _{q_+}$	$\ E\ _{q_+}$
K_1	$< 212/100$	$< 1/10^{16}$
K_2	$< 343/100$	$< 1/10^8$

They are finite rational inequalities for the recurrence just displayed: all coefficients of E through degree 80 vanish and only the next e coefficients enter its norm. In particular $\|E\|_{q_+} < 1$, so the identity $K^{-1} = V \sum_{j \geq 0} E^j$ proves, including its infinite tail,

$$\|K_1^{-1}\|_q < 7/2, \quad \|K_2^{-1}\|_q < 7/2.$$

Also $\|K_1\|_q = 2$ and $\|K_2\|_q = 4 - 2q < 14/5$. Combining these with (6.10) proves (6.9). $\square$

Lemma 6.7 (A finite numerator bound and signed conversion). *Suppose $Z_{m,r,p}(q) \leq B(r, p)$ for every $m > p$. Then, with $N = 3d^2 + 7p + 6$,*

$$(6.11) \quad \|H_{r,p}\|_q \leq 420NB(r, p),$$

and, for every $g \geq 0$,

$$(6.12) \quad [z^g]F_{r,p} \leq \frac{9}{4}\|H_{r,p}\|_q F_g.$$

Proof. Only coefficients through degree N of $F_{r,p}$ can enter the polynomial $A_{r,p} = B_d(1 - z - z^2)F_{r,p}$. Positivity of each graph polynomial and $g \geq m + 5$ imply

$$\sum_{j=0}^N [z^j]F_{r,p} q^j \leq NB(r, p),$$

since fewer than N multiplicities contribute. Using $\|1 - z - z^2\|_q = 2$ and (6.9) gives $\|A_{r,p}\|_q \leq 20NB(r, p)$ and then (6.11). This step never substitutes the potentially divergent value $F_{r,p}(q)$ for its finite truncation.

Write $H_{r,p} = \sum_j h_j z^j$. Its support starts at degree at least eight. From $(1 - z - z^2)^{-1} = \sum_{n \geq 0} F_{n+1} z^n$,

$$[z^g]F_{r,p} = \sum_{j=8}^g h_j F_{g-j+1}.$$

For $8 \leq j \leq g$, Binet's formula and $F_n \leq q^{1-n}$ give

$$\frac{F_{g-j+1}}{F_g} \leq \frac{\sqrt{5}}{1-q^{16}} q^j < \frac{9}{4} q^j.$$

For the last strict inequality it suffices to square and check $5 < [(9/4)(1-q_+^{16})]^2$. Take absolute values of the h_j . For $g < 8$ the raw coefficient is zero, which also handles $g = 0$. $\square$

Remark 6.8. The signed argument is necessary for this correction. For $r = 1, p = 2$, the word $(4, 4)$ contributes at genus eight, whereas no raw word for these pivots has genus nine. At multiplicity four the remaining coordinate has a carry self-pair and is at least two; multiplicities at least five have genus at least ten. Hence $[z^9](1-z-z^2)F_{1,2} = -1$.

6.4. Strong runs at the Fibonacci point. For the remainder of the proof all matrices are evaluated at q . On four coordinates let

$$D_{ij} = q^{(i+j)/2} \mathbf{1}_{i+j \geq 4}, \quad C_{ij} = q^{(i+j)/2} \mathbf{1}_{i+j \geq 3} \mathbf{1}_{i,j \leq 3}.$$

Let E be D with its fourth row and column set to zero. Thus E is the alphabet-three strong transfer. Put

$$(6.13) \quad R = \frac{917}{1000}, \quad k_0 = \frac{10003}{10000}, \quad \chi = \frac{47597}{50000}, \quad t = \frac{95761}{100000},$$

$$u_* = (1, 13/50)^T, \quad v_* = (823/1250, 97/625)^T, \quad \nu = v_*^T u_* = \frac{10918}{15625}.$$

Lemma 6.9 (Uniform transfer block). *Let $Q = (v, w)$ be an orthonormal basis for the nonzero eigenspaces of C , in the order $1, \mu = -q^2$. For all $k, n \geq 1$,*

$$(6.14) \quad \left| Q^T D^k Q \begin{pmatrix} 1 & 0 \\ 0 & \mu^n \end{pmatrix} \right| \leq R^{\max(k-3, 0)} u_* v_*^T$$

entrywise. Moreover $\|D\|_2 < R$, $\|E\|_2 < 19/25$, and $Q^T E Q = Q^T D Q$.

Proof. The nonsymmetric matrix similar to the cap- a restriction of D has entries $q^j \mathbf{1}_{i+j \geq 4}$. Its products with the positive vectors $(10000, 17164, 23907, 23907)^T$ for $a = 4$ and $(1, 2, 3)^T$ for $a = 3$ are componentwise strictly smaller than R and $19/25$ times those vectors, respectively. Each comparison reduces to $q^2 = 1 - q$. The positive-vector bound on spectral radius, and symmetry of the original matrices, give the asserted operator norms.

One may choose

$$v = \frac{(q, \sqrt{q}, q, 0)^T}{\sqrt{1+q^2}}, \quad w = \frac{(1, -q^{3/2}, -q^2, 0)^T}{\sqrt{1+q^2}}.$$

Their required moments, writing $\alpha_k = v^T D^k v$, $o_k = v^T D^k w$, and $\beta_k = w^T D^k w$, are

$$\begin{aligned} \alpha_1 &= \alpha_2 = (7 - 6q)/5, & o_1 &= (1 - 3q)/5, & o_2 &= (11 - 18q)/5, \\ \beta_1 &= (-7 + 11q)/5, & \beta_2 &= (3 - 4q)/5, & \beta_4 &= (-181 + 293q)/5. \end{aligned}$$

These exact identities give the rational bounds

$$\begin{aligned} \alpha_1, \alpha_2 &< A = \frac{823}{1250}, & |o_1| &< \frac{171}{1000}, & |o_2| &< \frac{1}{40}, \\ |\beta_1| &< \frac{41}{1000}, & \beta_2 &< B = \frac{66}{625}, & \beta_4 &< J = \frac{17}{1000}. \end{aligned}$$

Also $|\mu| < \rho = 191/500$. For $k \geq 3$, Cauchy-Schwarz gives

$$|\alpha_k| \leq R^{k-2} A, \quad |o_k| \leq R^{k-3} \sqrt{AJ}, \quad |\beta_3| \leq \sqrt{BJ}, \quad |\beta_k| \leq R^{k-4} J \quad (k \geq 4).$$

For example, the middle bound follows from $(Dv)^T D^{k-3}(D^2w)$. To check all powers, write $M_* = u_* v_*^T$. For $k = 1, 2$ use the listed moments, multiplying the second column by at most ρ . For $k \geq 3$ the sufficient rational comparisons are

$$RA < M_{*,11}, \quad AJ < M_{*,21}^2, \quad \rho^2 AJ < M_{*,12}^2, \quad \rho^2 BJ < M_{*,22}^2, \quad \rho J/R < M_{*,22}.$$

Direct substitution proves them. These inequalities prove (6.14) for every power, rather than only for a finite list of lengths. The fourth coordinates of v, w vanish, so $Q^T EQ = Q^T DQ$. $\square$

Proposition 6.10 (Uniform graph-value estimate). *Define*

$$e_D = \left\lfloor \frac{r-1}{2} \right\rfloor + \left\lfloor \frac{p-1}{2} \right\rfloor - \mathbf{1}_{p \neq 2r}, \quad e_E = \max \left(0, \left\lfloor \frac{p-1}{2} \right\rfloor - r \right),$$

and $s_4(z) = z + z^2 + z^3 + z^4$. For every $m > p$,

$$(6.15) \quad Z_{m,r,p}(q) \leq k_0^3 q^8 s_4(q) R^{e_D+2e_E} \chi^{d-16}.$$

Proof. Call a maximal nonempty sequence of ordinary strong edges in a component of the reflection graph a *strong run*. We first count its edges and boundaries, then estimate its transfer product.

There are $\lfloor (r-1)/2 \rfloor$ ordinary strong pairs targeting r and $\lfloor (p-1)/2 \rfloor$ targeting p . The only such pair incident to a pivot is $\{r, p-r\}$, unless $p = 2r$, when this is a loop at r . Deleting the pivots therefore leaves exactly e_D ordinary strong edges. Among these, a pair with both endpoints greater than r must target p and is uniquely specified by

$$r < i < p - i, \quad i = r + 1, \dots, \left\lfloor \frac{p-1}{2} \right\rfloor.$$

There are exactly e_E such edges. Neither endpoint can have a strong edge targeting r , so each of these is an entire one-edge strong run. It uses the alphabet-three matrix E . Every other run of k edges is bounded by D^k , enlarging any smaller vertex alphabet as needed. Every carry endpoint is greater than its target and has alphabet three; the zero fourth row and column of C enforce this alphabet at each strong/carry junction.

For the boundary count, each nonpivot coordinate has one reflection incidence for each target. Such an incidence is either an ordinary edge, a loop, or an edge lost on deleting a pivot. Pivot r has only the possible nonloop neighbor $p-r$, and pivot p only the possible nonloop neighbor $m+r-p$; the pivots are not adjacent. Hence there are at most two free boundary incidences. For each target t , loops can occur only at $2i = t$ or $2i = m+t$, so there are at most four loop incidences in total. If P is the number of path components, F the number of free incidences, and L the number of loop incidences, then

$$2P = F + L, \quad F \leq 2, \quad L \leq 4, \quad P \leq 3.$$

An isolated vertex counts as a zero-edge path with two boundary incidences, so it is included in this identity.

Let $h_4 = (q^{1/2}, q, q^{3/2}, q^2)^T$ and let $h_{4,+}$ be the same vector with its first entry zero. Free endpoint vectors are bounded by h_4 , and loop endpoint vectors by $h_{4,+}$, because either loop threshold forces a value at least two. Thus all boundary norms combined cost at most

$$(6.16) \quad \|h_4\|_2^F \|h_{4,+}\|_2^L = s_4(q)^{F/2} (s_4(q) - q)^{L/2} \leq s_4(q),$$

since $s_4(q) > 1$ and $s_4(q) - q < 1$. The same formula handles the two restrictions on an isolated vertex through the inner product of its two boundary vectors.

Every ordinary cycle contains both strong and carry edges. Indeed, target labels alternate around a cycle. In an all-strong cycle the two target reflections compose to a translation by d or $-d$; in an all-carry cycle they do the same. Successive pairs of edges use the same ordering of target labels, so iterating that nonzero translation cannot close the cycle.

For each cycle, rotate the transfer product to alternate nonempty strong and carry runs. Write $T_i = D^{k_i}$ for an ordinary strong run and $T_i = E$ for an alphabet-three strong run, and put $\Lambda_n = \text{diag}(1, \mu^n)$. Its partition is bounded by

$$\text{tr}(T_1 C^{m_1} \cdots T_j C^{m_j}) = \text{tr} \prod_{i=1}^j (Q^T T_i Q \Lambda_{n_i}), \quad n_i \geq 1.$$

By Lemma 6.9, the absolute value of each block is bounded entrywise by $a_i M_*$, where $M_* = u_* v_*^T$ and

$$a_i = R^{\max(k_i-3, 0)} \quad (T_i = D^{k_i}), \quad a_i = 1 \quad (T_i = E).$$

The latter follows from $Q^T E Q = Q^T D Q$. Since $M_*^j = \nu^{j-1} M_*$, taking absolute values and multiplying the nonnegative comparison matrices gives the cycle bound

$$\left(\prod_i a_i \right) \text{tr}(M_*^j) = \left(\prod_i a_i \right) \nu^j.$$

In particular, no dimension factor is charged for a cycle.

For a path, remove the strong run at either endpoint if present and charge its operator norm. A path consisting entirely of strong edges has just one such run. We call these runs *unpaid* only with respect to the additional factor ν/R^3 below: their edges are already charged. At most two runs per path are unpaid. Unless the whole path has been removed, what remains begins and ends with carry runs. If it contains $j \geq 1$ strong runs, its product after projection into the carry image has the form

$$\Lambda_{n_0} (Q^T T_1 Q \Lambda_{n_1}) \cdots (Q^T T_j Q \Lambda_{n_j}), \quad n_i \geq 1.$$

Orthogonal projection and Λ_{n_0} have operator norm at most one. Taking entrywise absolute values preserves the Euclidean norm of each endpoint vector. The same block comparison consequently bounds this middle product by

$$\left(\prod_i a_i \right) \|M_*^j\|_2 = \left(\prod_i a_i \right) \kappa \nu^j, \quad \kappa = \frac{\|u_*\|_2 \|v_*\|_2}{\nu} < k_0.$$

If there is no remaining strong run, the carry product has norm at most one. This includes a zero-edge remainder; a path removed in its entirety was already bounded by its endpoint norms and strong-edge charges. Thus in every case one factor k_0 per path suffices, at most k_0^3 in total. We call all strong runs in cycles and all nonendpoint strong runs in paths *paid*.

To count paid runs, consider the strong graph before deleting r . On its vertices $1, \dots, p-1$, the substitution $x = p-i$ turns the reflections into $p-x$ and $p+d-x$, with ports $1, \dots, d$. The residue-class path argument in Section 6.2 applies here too: it only requires each residue class to be nonempty, which holds because $p-1 \geq d$. If f_s classes are fixed by the residue involution, then $f_s \leq 2$, because $2a \equiv p \pmod{d}$ has at most two solutions. The $(d-f_s)/2 \geq d/2-1$ paired-class components each contain an ordinary strong edge, even if their classes have only one vertex each. The deleted pivot r is port d . Removing that endpoint can destroy at most one of these components and cannot split another. At least $d/2-2$ nonempty strong components therefore survive. These components are precisely strong runs after carry edges are restored. At most $2P \leq 6$ runs were declared unpaid, so the total number j of paid runs satisfies

$$(6.17) \quad j \geq d/2 - 8.$$

This argument involves only the strong interval and the boundary count; in particular it applies throughout $p < m \leq 2p$ as well as for $m > 2p$.

The constants satisfy the exact rational comparisons

$$\frac{\nu}{R^3} < \chi^2 < 1, \quad \frac{\|u_*\|_2 \|v_*\|_2}{\nu} < k_0, \quad \frac{19}{25} < R^3, \quad \chi < t, \quad R < t^2.$$

Charge R per ordinary strong edge outside the E runs and R^3 per E edge. An unpaid run is covered by its operator norm: $\|D^k\|_2 \leq R^k$ or $\|E\|_2 < 19/25 < R^3$. For a paid run, the preceding product estimates give

$$R^{\max(k-3,0)} \nu \leq R^k \frac{\nu}{R^3} \quad \text{for a } D \text{ run}, \quad \nu = R^3 \frac{\nu}{R^3} \quad \text{for an } E \text{ run}.$$

Every strong edge is counted exactly once, so the total strong charge is $R^{e_D - e_E + 3e_E} = R^{e_D + 2e_E}$. By (6.17), the extra factor is bounded by

$$(\nu/R^3)^j \leq \chi^{2j} \leq \chi^{d-16}.$$

When $d < 16$, the last negative exponent simply weakens the bound. Multiplying this estimate by the path factor k_0^3 , the global boundary bound (6.16), and the pivot weight q^8 proves (6.15). $\square$

6.5. Summation over omitted pivots.

Proof of Theorem 6.2. Combining Lemma 6.7 with Proposition 6.10 gives the coefficient multiplier

$$945(3d^2 + 7p + 6)k_0^3 q^8 s_4(q) R^{e_D + 2e_E} \chi^{p-r-16}.$$

Replace d^2 by p^2 . Put $J = \lfloor (p-1)/2 \rfloor$ and $a = R^2 \chi/t < 1$. Since $R^{\lfloor (r-1)/2 \rfloor} \leq t^{r-2}$, the region $r \leq J$ is bounded, after reversing its geometric sum, by

$$\sum_{r \leq J} R^{e_D + 2e_E} \chi^{p-r} \leq \frac{\chi^{p-J} (Rt)^J}{Rt^2(1 - R^2 \chi/t)}.$$

Here replacing the pivot-deletion indicator by one only enlarges the bound. For $r > J$ there are no E edges; summing backwards from $r = p-1$ gives

$$\sum_{J < r < p} R^{e_D} \chi^{p-r} \leq \frac{R^J t^p}{Rt^2(t/\chi - 1)}.$$

An empty region is harmless because the displayed upper bound is positive.

For a quadratic $P(k) = ak^2 + bk + c$, define

$$(6.18) \quad \mathcal{T}_x(K; P) = x^K \left[\frac{aK^2 + bK + c}{1-x} + \frac{(2aK + b)x}{(1-x)^2} + \frac{ax(1+x)}{(1-x)^3} \right] = \sum_{k \geq K} P(k)x^k.$$

Set

$$X = Rt\chi, \quad Y = Rt^2, \quad U = 1 - R^2\chi/t, \quad V = t/\chi - 1,$$

and

$$P_o(k) = 12k^2 + 26k + 16, \quad P_e(k) = 12k^2 + 38k + 32.$$

These are $3p^2 + 7p + 6$ for $p = 2k+1, 2k+2$, respectively. In both cases $J = k$. Pairing these two parities, and replacing q by q_+ in the positive prefactor, bounds the multiplier for every pivot $p > 2K$ by the following explicit rational number:

$$(6.19) \quad \mathcal{B}_K = \frac{945q_+^8 s_4(q_+) k_0^3 \chi^{-16}}{Rt^2} \left[\frac{\mathcal{T}_X(K; \chi P_o + \chi^2 P_e)}{U} + \frac{\mathcal{T}_Y(K; tP_o + t^2 P_e)}{V} \right].$$

All entries in this expression are the specified rational constants; $0 < X < Y < 1$ and $U, V > 0$. Substitution in (6.18) and integer cross-multiplication give the exact inequality

$$\mathcal{B}_{150} < \frac{221714122}{10^9} < \frac{221715}{10^6}.$$

This finite arithmetic comparison bounds the whole infinite geometric sum in (6.19). Together with the preceding structural lemmas it proves (6.1) for every genus. For a fixed genus the pivot sum itself is finite by Lemma 6.3; the displayed majorant additionally converges absolutely. $\square$

Corollary 6.11 (Integer envelope). *Let $C_g(300) = \sum_{2 \leq p \leq 300} \sum_{r < p} [z^g] F_{r,p}$ be the exact coefficient of the raw cap-four prefix. Then the full raw cap-four coefficient is at most*

$$C_g(300) + \mathbf{1}_{g \geq 604} \left\lfloor \frac{221715 F_g}{1000000} \right\rfloor.$$

Proof. The omitted coefficient is an integer, so the floor of its real upper bound is still an upper bound. Its support starts at genus 604. $\square$

7. CONVERGENT MASSES OF THE GENUINE RESIDUAL POPULATIONS

Let A_g count genuine depth-four Kunz words of genus g with at least two coordinates equal to four, and let B_g count genuine depth-five words with at least two coordinates equal to five. Write $A(z) = \sum_g A_g z^g$ and $B(z) = \sum_g B_g z^g$. We prove

$$(7.1) \quad A(1247/2000) < 2024, \quad B(633/1000) < 113.$$

These are inequalities at specified positive real arguments. In particular, the first inequality does not assert convergence there of the unrestricted cap-four relaxation. The use of weighted Kunz graphs and transfer matrices continues methods of Zhu [14] and Bacher [1]; the additional self-pair restrictions below are essential to the present explicit estimate.

7.1. The multiple-five mass. For $t \in \{4, 5\}$ let

$$D_t(z)_{ij} = z^{(i+j)/2} \mathbf{1}_{i+j \geq t}, \quad 1 \leq i, j \leq t,$$

and put $s_5 = \sum_{j=1}^5 z^j$, $h_5 = z^3 + z^4 + z^5$ and $h_4 = z^2 + z^3 + z^4$. The Schatten fourth norm of a real symmetric matrix M is $(\text{tr } M^4)^{1/4}$.

Lemma 7.1. *Suppose $0 < a_5 < b_4 < 1$, $\|D_5(z)\|_4 \leq a_5$, $\|D_4(z)\|_4 \leq b_4$, $s_5 \geq b_4$, $h_5 \leq a_5$ and $h_4 \leq b_4$. Set $k = (a_5/b_4)^{1/2}$. Then*

$$B(z) \leq \frac{z^{10} (s_5/b_4) k^{-1}}{(1 - b_4)(1 - b_4 k)(1 - a_5)}.$$

In particular $B(633/1000) < 113$.

Proof. Fix the two rightmost fives at $r < p < m$, delete these fixed vertices, and retain the two reflection constraints. Carry endpoints have alphabet $\{1, 2, 3, 4\}$; enlarging the strong-endpoint alphabet to $\{1, 2, 3, 4, 5\}$ can only increase the partition. The ordinary graph is a union of paths and even cycles, the latter having length at least four. Regard an isolated vertex as a path with two endpoint incidences. Let P be the number of paths, u the number of free incidences, L_5, L_4 the numbers of strong and carry loop incidences, and S, C the numbers of ordinary strong and carry edges. Reflection counting gives

$$2P = u + L_5 + L_4, \quad S + C = m - 3 - P, \quad u \leq 2, \quad 2S + L_5 \geq r + p - 4.$$

The final inequality includes $p = 2r$: deleting the second pivot then removes a strong loop rather than an ordinary strong edge. Cauchy-Schwarz on each path bounds its endpoint factors by $s_5^{u/2} a_5^{L_5/2} b_4^{L_4/2}$ and its edge factors by $a_5^S b_4^C$. For a cycle of length $\ell \geq 4$, Schatten Hölder and $\|M\|_\ell \leq \|M\|_4$ give the same edge bound with no dimension factor. Alphabet projections are contractions and preserve these bounds. Consequently the nonpivot partition is at most

$$b_4^{m-3} (s_5/b_4)^{u/2} k^{2S+L_5} \leq b_4^{m-3} (s_5/b_4) k^{r+p-4}.$$

Now put $r = 1 + i$, $p = 2 + i + j$, $m = 3 + i + j + \ell$, with $i, j, \ell \geq 0$, and sum the three geometric series.

At $z = 633/1000$ take

$$a_5 = \frac{876716}{10^6}, \quad b_4 = \frac{982645}{10^6}, \quad \tau = \frac{928171}{10^6}.$$

Every norm comparison is an exact rational inequality, since

$$\text{tr } D_t(z)^4 = \sum_{i_1, \dots, i_4=1}^t z^{i_1+i_2+i_3+i_4} \prod_{j=1}^4 \mathbf{1}_{i_j+i_{j+1} \geq t}, \quad i_5 = i_1.$$

The two trace sums are strictly below a_5^4, b_4^4 , respectively; the endpoint inequalities hold, and $1/2 < \sqrt{a_5 b_4} < \tau < 1$. The preceding bound equals $z^{10} s_5 / [\sqrt{a_5 b_4} (1 - \sqrt{a_5 b_4}) (1 - a_5) (1 - b_4)]$. The function $1/[x(1-x)]$ increases for $1/2 < x < 1$, so replacing the square root by τ gives the rational upper bound

$$\frac{z^{10} s_5}{\tau(1-\tau)(1-a_5)(1-b_4)} < 112.194409824952 < 113.$$

All summations can first be made over finite boxes; monotone convergence then proves convergence as well as the stated bound. $\square$

7.2. Conditional self-pair contraction. Write $Z_{m,r,p}(z)$ for the raw cap-four partition with its two rightmost fours fixed at $r < p$, including their factor z^8 . Let $Z_{m,r,p}^{\text{self}}(z)$ be the subpartition avoiding

$$F_i = \{x_i = 1, x_{2i} = 3\}, \quad p < i < m/2.$$

Every genuine word belongs to this subpartition, by the noncarry Kunz inequality at (i, i) . For $0 < z \leq 1$ put

$$a = z, \quad b = z^2 + z^3, \quad s = a + b, \quad \epsilon = \frac{abz^3}{s(ab + s^2)}, \quad \delta = \frac{\epsilon}{(1+z)^2},$$

$$\beta = 1 - \epsilon, \quad \gamma = 1 - \delta, \quad T = \gamma/\beta, \quad d = p - r, \quad N = \max\{0, \lfloor (m-1)/2 \rfloor - p\}.$$

Lemma 7.2. *For every pivot pair and multiplicity,*

$$(7.2) \quad Z_{m,r,p}^{\text{self}}(z) \leq \beta^N T^{1+\lfloor d/3 \rfloor} Z_{m,r,p}(z).$$

For $m \leq 2p$ the stronger bound $Z_{m,r,p}^{\text{self}} \leq Z_{m,r,p}$ will be used.

Proof. Assume $m > 2p$, and let

$$V = \{p+1, \dots, \lfloor (m-1)/2 \rfloor\}, \quad |V| = N.$$

Give the raw words the probability measure proportional to their weights. For a center $i \in V$, write

$$\ell_r(i) = m + r - i, \quad \ell_p(i) = m + p - i, \quad B_i = \{i, 2i, \ell_r(i), \ell_p(i)\}.$$

Every coordinate in B_i exceeds p , so its alphabet is $\{1, 2, 3\}$. Both leaves $\ell_r(i), \ell_p(i)$ exceed $m/2$. The center has precisely these two retained neighbors, and has no loop. All retained constraints touching the gadget are carry constraints of threshold three. The only possible identification in B_i is $2i = \ell_r(i)$ or $2i = \ell_p(i)$.

Local failure families. First suppose the four vertices are distinct. Fix the coordinates outside B_i to an assignment admitting at least one completion. The family

$$x_i = 1, \quad x_{2i} = 3, \quad x_{\ell_r(i)}, x_{\ell_p(i)} \in \{2, 3\}$$

has total gadget weight $ab^2 z^3$. It satisfies all retained constraints for every such exterior assignment: the center's constraints are satisfied by its leaves, and every other gadget spin is at least

two. Dropping all restrictions except the two center–leaf edges bounds the whole gadget partition by

$$s(ab^2 + bs^2).$$

Indeed center spin one contributes ab^2 , center spin two or three contributes at most bs^2 , and the target contributes at most s . The ratio of the displayed numerator to this denominator is ϵ . If the target coincides with one leaf, the corresponding numerator and denominator are abz^3 and $ab^2 + bs^2$. Their ratio is

$$\frac{az^3}{ab + s^2} > \epsilon.$$

These bounds also allow any additional edge or loop among the gadget vertices, since the failure families satisfy it and it only reduces the denominator.

An ordering of the centers. For distinct centers draw $i \rightarrow j$ when

$$(7.3) \quad 2j \in \{\ell_r(i), \ell_p(i)\}.$$

This edge means that F_i should be processed before F_j to retain both choices at every reflection leaf. We construct a set of postponed centers of size at most $1 + \lfloor d/3 \rfloor$, and an order of all other centers in which every arrow goes to a later or postponed center. Put

$$c = (2m + r + p)/6, \quad I = [c - d/2, c + d/2].$$

Process centers outside I in decreasing order of $|i - c|$, with arbitrary tie breaking. For an arrow $i \rightarrow j$, write $j = (m + t - i)/2$ with $t \in \{r, p\}$. Since $m + t - 3c = \pm d/2$,

$$|j - c| \leq |i - c|/2 + d/4 < |i - c| \quad (i \notin I).$$

Thus an outside center never points to an earlier center. The weak version of the same inequality shows that both affine maps preserve I , so no center in I points to an outside center.

The remaining centers form the consecutive integer set $V \cap I$, with at most $d + 1$ elements. If there are $d + 1$ elements, postpone one endpoint. The remaining set has at most d elements and diameter strictly less than d . For any fixed target vertex j , its two possible incoming sources $m + r - 2j$ and $m + p - 2j$ differ by d ; therefore its indegree in the remaining graph is at most one. Every nontrivial strongly connected component is consequently a simple directed cycle: strong connectivity supplies an internal incoming edge at every vertex, leaving exactly one internal incoming and outgoing edge at each vertex. A two-cycle would give

$$i + 2j = m + t_1, \quad j + 2i = m + t_2, \quad i - j = t_2 - t_1 \in \{-d, 0, d\},$$

which is impossible for distinct vertices of this set. Self-edges were excluded in (7.3). Every directed cycle thus has at least three vertices. Postpone one vertex from each cyclic component and put the remaining acyclic graph in topological order, with i before j on every edge $i \rightarrow j$. The postponed set has size

$$f \leq 1 + \lfloor d/3 \rfloor.$$

All centers not postponed have now been ordered; process the postponed ones last in any order.

Survival under the preceding avoidances. Consider a nonpostponed center i , and condition on avoidance of all preceding F_j . On a fixed exterior assignment outside B_i with positive mass under that conditioning, we claim that the entire local failure family above still survives. A preceding event disjoint from B_i is already avoided by the fixed exterior assignment. For one meeting B_i , the possible intersections are exhaustive as follows. An earlier center cannot be a reflection leaf, since centers are below $m/2$ and leaves above it. If $j = 2i$, its center has spin three, so F_j is false. If $2j = i$, its target has spin one, so F_j is false. Equal centers or equal doubled targets force $j = i$. The only remaining possibility is a preceding target at a reflection leaf, precisely an arrow $i \rightarrow j$, which the ordering excludes. In a three-vertex gadget, a target at the shared leaf $2i$ would again force $j = i$.

It follows that the conditional numerator on this exterior slice is at least the full weight already displayed. The conditional denominator is at most the unrestricted gadget denominator displayed there. The exterior weight cancels. Averaging over the exterior slices proves

$$\Pr(F_i \mid \text{all preceding avoidances}) \geq \epsilon.$$

This argument does not require preceding avoidances to be measurable outside B_i ; their survival was checked on each proposed completion.

For a postponed center use the single assignment

$$x_i = 1, \quad x_{2i} = 3, \quad x_{\ell_t(i)} = 2 \quad \text{when } \ell_t(i) \neq 2i.$$

It also makes false every preceding event meeting the gadget, including an event whose target is a remaining leaf, now of spin two. Its weight in the four-vertex case is z^8 , giving conditional probability at least

$$\frac{z^8}{s(ab^2 + bs^2)} = \delta.$$

In the three-vertex case its weight is z^6 , and the bound is larger because $s > z^2$. The same slice argument therefore applies in any order to the postponed centers. Multiplying the conditional avoidance bounds gives

$$Z_{m,r,p}^{\text{self}} \leq \beta^{N-f} \gamma^f Z_{m,r,p} \leq \beta^N T^{1+[d/3]} Z_{m,r,p}.$$

If any conditioning event has zero mass, the final left side is zero and the assertion follows immediately. For $m \leq 2p$ there are no eligible centers, and the raw bound suffices. $\square$

7.3. A uniform envelope for the absolute spectral expansion. From now on set $z = z_4 = 1247/2000$, and define

$$D_{ij} = z^{(i+j)/2} \mathbf{1}_{i+j \geq 4}, \quad 1 \leq i, j \leq 4, \\ C_{ij} = z^{(i+j)/2} \mathbf{1}_{i+j \geq 3} \mathbf{1}_{i,j \leq 3}, \quad J_3 = \text{diag}(1, 1, 1, 0), \quad E = J_3 D J_3.$$

The nonzero carry eigenvalues are $\lambda > 0 > \mu$, where $\lambda^2 - b\lambda - ab = 0$ and $\mu = b - \lambda$. Let $Q = (q_+, q_-)$ have orthonormal eigenvectors as columns, with q_+ nonnegative, and put $\rho = |\mu|/\lambda$. Thus

$$C^n = \lambda^n q_+ q_+^T + \mu^n q_- q_-^T \quad (n \geq 1).$$

We specify the spectral partitions used below. For a fixed reflection graph with $m > 2p$, retain each maximal strong run as an exact matrix product, including its endpoint alphabet projections, and expand every maximal carry run by the displayed identity. A mode assignment chooses $+$ or $-$ for each carry run. After all strong blocks and boundaries have been contracted, let W_ω denote its scalar weight, including z^8 and any component having no carry edge. Define

$$(7.4) \quad P_{m,r,p} = W_{(+, \dots, +)}, \quad A_{m,r,p}^{\text{abs}} = \sum_{\omega} |W_\omega|, \quad M_{m,r,p} = A_{m,r,p}^{\text{abs}} - P_{m,r,p}.$$

The positive mode has nonnegative endpoint and strong-block factors, so $P_{m,r,p} \geq 0$. Taking absolute values of the projected blocks and endpoint factors before multiplying gives exactly $A_{m,r,p}^{\text{abs}}$. In particular

$$(7.5) \quad 0 \leq P_{m,r,p} \leq A_{m,r,p}^{\text{abs}}, \quad 0 \leq Z_{m,r,p} \leq P_{m,r,p} + M_{m,r,p} = A_{m,r,p}^{\text{abs}}.$$

Here the *mixed* remainder $M_{m,r,p}$ includes every assignment containing a negative mode, including the all-negative assignment. This convention differs from the two-pure-mode decomposition used for algebraic cancellation in Section 6.2.

There is one carry component of zero ordinary-edge length that must still receive two modes. It occurs at $m = 3p - 2r > 2p$, when deletion of port p leaves the isolated vertex $i = 2p - r$ with a carry loop. Its free and loop boundary vectors are

$$h = (z^{1/2}, z, z^{3/2}, 0)^T, \quad h_+ = (0, z, z^{3/2}, 0)^T.$$

Both lie in $\text{im } C$, so its scalar weight is split as

$$(7.6) \quad h^T h_+ = (h^T q_+)(q_+^T h_+) + (h^T q_-)(q_-^T h_+).$$

This is the convention for W_ω at that corner; its absolute mode sum is at most $\|h\|_2 \|h_+\|_2$ by Cauchy–Schwarz. An isolated vertex in the strong core is kept as its ordinary positive scalar and is not given this artificial carry expansion.

The constants used below are

$$(7.7) \quad R = \frac{23389}{25000}, \quad S = \frac{72789}{100000}, \quad L = \frac{101778011}{10^8}, \quad u = \begin{pmatrix} 1 \\ 7/25 \end{pmatrix}, \quad v = \begin{pmatrix} 47/50 \\ 9/50 \end{pmatrix}.$$

Lemma 7.3. *We have $\|D\|_2 < R$, $\|E\|_2 < S$, $1 < \lambda < L$ and $\rho < 39/100$. For every $k, n \geq 1$, entrywise,*

$$(7.8) \quad \left| Q^T D^k Q \text{diag}(\lambda^n, \mu^n) \right| \leq R^k \lambda^n u v^T.$$

For a single restricted strong edge and every $n \geq 1$,

$$\left| Q^T E Q \text{diag}(\lambda^n, \mu^n) \right| \leq S \lambda^n u v^T.$$

Proof. Conjugating D by $H = \text{diag}(z^{j/2})_{j=1}^4$ gives $M := H^{-1} D H = (z^j \mathbf{1}_{i+j \geq 4})_{ij}$. The positive test vector $(1000000, 1710962, 2377416, 2377416)^T$ satisfies $Mx < Rx$ componentwise. For the corresponding three-dimensional restriction M_3 , $(1000000, 2146345, 3002947)^T$ satisfies $M_3 x < Sx$. Perron–Frobenius and symmetry prove the norm bounds. The quadratic equation gives the stated λ, ρ inequalities by rational squaring.

Write $\alpha_k = q_+^T D^k q_+$, $\omega_k = q_+^T D^k q_-$ and $\nu_k = q_-^T D^k q_-$. The following rational strict upper bounds on absolute values suffice:

k	$ \alpha_k $	$ \omega_k $	$ \nu_k $
1	68/100	18/100	5/100
2	69/100	3/100	11/100
4	not needed	not needed	2/100

They can be checked without numerical eigenvectors. Set

$$y_+ = (b/\lambda, 1, 1, 0)^T, \quad y_- = (\lambda/a, -1, -1, 0)^T, \\ \kappa_+ = a(b/\lambda)^2 + b, \quad \kappa_- = \lambda^2/a + b.$$

Then $q_\pm = H y_\pm / \sqrt{\kappa_\pm}$. Substitute the positive root in these finite expressions, enclose its square root by rational squares, and compare; the denominator signs are positive. This also gives the same first-power moments for E , because $J_3 q_\pm = q_\pm$.

For $k \geq 2$, Cauchy–Schwarz gives $|\alpha_k|/R^k \leq (69/100)/R^2$. For $k \geq 3$ it gives

$$\frac{|\omega_k|}{R^k} \leq \frac{\sqrt{(69/100)(2/100)}}{R^3}, \quad \frac{|\nu_3|}{R^3} \leq \frac{\sqrt{(11/100)(2/100)}}{R^3},$$

and for $k \geq 4$, $|\nu_k|/R^k \leq (2/100)/R^4$. Indeed apply D to q_+ and D^2 to q_- for the off-diagonal estimate, and D^2 to both copies of q_- for the last estimate. For $k = 1, 2$ use the table directly. The second column in (7.8), after division by λ^n , gains $\rho^n \leq 39/100$. Comparison with the four entries of $u v^T$ proves the assertion; square the positive off-diagonal bounds if desired. The same comparisons with denominator S prove the restricted case. This extends the finite moment checks to every power. $\square$

The hypothesis $n \geq 1$ is essential: the assertion is about a strong run followed by a positive-length carry run. For example, $\nu_2/R^2 > 3/25 > (uv^T)_{22}$, so a bare strong-run assertion would be false. The zero-edge component in (7.6) is handled by its boundary estimate; (7.8) is never used with $n = 0$.

Here are the endpoint and component consequences of the lemma. A vertex with index greater than r has at most one strong incidence, so every internal vertex of a strong run has the full alphabet $\{1, 2, 3, 4\}$. Alphabet restrictions on such a run therefore occur only at its endpoints. At a carry junction they do not change the projected block, since $J_3Q = Q$. Thus a run between carry components has the projected matrix $Q^T D^k Q$; if both endpoints of a strong edge exceed r , that edge is an entire one-edge strong run and the sharper E estimate applies. At a free or loop endpoint, absorb the alphabet and unary restrictions into the boundary vector. Strong prefixes and suffixes are then bounded by ordinary operator norms, using $\|E\|_2 < S$ for the restricted one-edge case. Alphabet projections and Q^T are contractions, while taking coordinatewise absolute values preserves Euclidean norm. Moreover

$$\|u\|_2^2 \|v\|_2^2 = \frac{77173}{78125} < 1, \quad v^T u = \frac{619}{625} < 1.$$

Every ordinary cycle has both a strong and a carry edge: two successive reflections in a cycle of either single kind translate an integer by the nonzero distance $p - r$, and cannot close. Thus cycles can be written as products of blocks with $k, n \geq 1$. Products introduce no extra path factor, and cycle traces are at most one after normalization. The at most two free endpoint incidences together contribute at most $s_4 = \sum_{j=1}^4 z^j$; loop endpoint norms are at most one, since $z^2 + z^3 + z^4 < 1 < s_4$. Counting strong edges gives

$$e_D = \lfloor (r-1)/2 \rfloor + \lfloor (p-1)/2 \rfloor - \mathbf{1}_{p \neq 2r}, \quad e_E = \max\{\lfloor (p-1)/2 \rfloor - r, 0\}.$$

With $\eta = R/L$ and $\sigma = S/R$, define

$$(7.9) \quad \mathcal{E}_{m,r,p} = z^8 s_4 L^{m-3} \eta^{e_D} \sigma^{e_E}.$$

The argument proves $Z_{m,r,p} \leq \mathcal{E}_{m,r,p}$ for every $m > p$, and $A_{m,r,p}^{\text{abs}} \leq \mathcal{E}_{m,r,p}$ for every $m > 2p$. For short multiplicities the ordinary graph estimate uses the same block bounds whenever there are carry edges and ordinary operator norms otherwise. The number of carry edges is at most $m - 3 - e_D$: the graph has $m - 3$ vertices and is a union of paths and cycles. The restricted-edge replacement supplies $(S/R)^{e_E}$, while $L > 1$ permits the stated larger carry exponent.

7.4. Finite bases and complete multiplicity tails. Write $f_d = 1 + \lfloor d/3 \rfloor$. Combining (7.2) with (7.5) gives the nonnegative decomposition that will be summed:

$$(7.10) \quad A(z) \leq \sum_{r < p} \sum_{m=p+1}^{2p} Z_{m,r,p} + \sum_{r < p} \sum_{m > 2p} T^{f_d} \beta^N (P_{m,r,p} + M_{m,r,p}).$$

It is enough initially to read this inequality over finite ranges; the bounds below establish convergence. We keep the exact factor β^N in this decomposition. For the finite bases and mixed tails we may enlarge it using the rational normalization

$$(7.11) \quad w = \frac{70651823329321825}{72057594037927936}, \quad w^2 \geq \beta, \quad w < 1, \quad J = (wL)^2 < 1.$$

Multiplying every vertex weight, including the two pivots, by w gives the normalized partitions

$$\hat{P}_{m,r,p} = w^{m-1} P_{m,r,p}, \quad \hat{A}_{m,r,p}^{\text{abs}} = w^{m-1} A_{m,r,p}^{\text{abs}}, \quad \hat{M}_{m,r,p} = w^{m-1} M_{m,r,p}.$$

For $m > 2p$ put

$$K_{m,r,p} = T^{f_d} w^{-2p - ((m-1) \bmod 2)}.$$

Since $m - 1 - 2p - ((m - 1) \bmod 2) = 2N$, for $X = P, M$ we have

$$(7.12) \quad T^{f_d} \beta^N X_{m,r,p} \leq K_{m,r,p} \widehat{X}_{m,r,p}.$$

A normalized block in (7.8) gains w^{k+n} on both sides; the boundary vectors each gain $w^{1/2}$ and the pivot factor gains w^2 . In particular the normalized zero-edge component is $wh^T h_+$, with both terms of (7.6) multiplied by w .

The carry-port topology of Section 6.2 identifies the carry components under $m \mapsto m + 2d$: each paired-port component gains four ordinary edges and each fixed-port component gains two, including the component from which p was deleted. The strong blocks and boundary factors remain the same. Every assignment contributing to M uses a negative mode on at least one such component. Its absolute weight therefore gains a factor at most $(w\lambda)^{2d} \rho^2$ after normalization. Summing these termwise bounds proves

$$(7.13) \quad \widehat{M}_{m+2d,r,p} \leq (w\lambda)^{2d} \rho^2 \widehat{M}_{m,r,p}.$$

For the positive mode, the rank-one projector $q_+ q_+^T$ supplies the same vector q_+ at every surviving port. Contracting these vectors against the fixed strong core is independent of which ports were paired in the carry graph. The carry-loop endpoints each contribute $q_+^T h_+$, and the free endpoint left by deleting p contributes $h^T q_+$. The number of carry loops depends only on parity, and the total carry exponent increases by two under $m \mapsto m + 2$. The zero-edge split in (7.6) has exactly these factors. Consequently

$$(7.14) \quad \widehat{P}_{m+2,r,p} = (w\lambda)^2 \widehat{P}_{m,r,p} \quad (m > 2p).$$

Thus the two bases $2p+1, 2p+2$ suffice for the pure sum, whereas the $2d$ bases $2p+s, 1 \leq s \leq 2d$, suffice for the mixed bound. The multiplier $K_{m,r,p}$ is constant on each progression being used.

Here are explicit definitions of the three finite pivot sums. For a positive integer cutoff P_0 , set

$$(7.15) \quad S(P_0) = \sum_{p=2}^{P_0} \sum_{r=1}^{p-1} \sum_{m=p+1}^{2p} Z_{m,r,p},$$

$$(7.16) \quad P(P_0) = \sum_{p=2}^{P_0} \sum_{r=1}^{p-1} \sum_{s=1}^2 \frac{K_{2p+s,r,p} \widehat{P}_{2p+s,r,p}}{1 - (w\lambda)^2},$$

$$(7.17) \quad M(P_0) = \sum_{p=2}^{P_0} \sum_{r=1}^{p-1} \sum_{s=1}^{2d} \frac{K_{2p+s,r,p} \widehat{M}_{2p+s,r,p}}{1 - (w\lambda)^{2d} \rho^2}.$$

All denominators are positive because $w\lambda < wL < 1$. Equations (7.12)–(7.14) show exactly how these finite sums bound the corresponding complete multiplicity sums in (7.10).

For evaluating them one may use the right eigenvectors $y_{\pm}$ from the proof of Lemma 7.3, with left vectors $z^j y_{\pm,j} / \kappa_{\pm}$. Their positive diagonal change of basis telescopes through complete products and endpoints and commutes with entrywise absolute values. It therefore yields the same pure and absolute-mode scalars just defined.

It remains to sum over pivots, rather than merely checking finitely many of them. Write

$$r = 2a + u_0, \quad p = 2b + v_0, \quad u_0, v_0 \in \{1, 2\}, \quad h = \begin{cases} 0, & u_0 < v_0, \\ 1, & u_0 \geq v_0. \end{cases}$$

The exact domain is $a \geq 0, b \geq a + h$. Also $e_D \geq a + b - 1$ and $e_E = \max\{b - 2a - u_0, 0\}$. Define

$$\mathcal{H}_{u_0 v_0}(A, B) = \sum_{a \geq 0} \sum_{b \geq a+h} A^a B^b \sigma^{\max\{b-2a-u_0, 0\}}.$$

Splitting at $b = 2a + u_0$ gives

$$(7.18) \quad \mathcal{H}_{u_0 v_0}(A, B) = \frac{B^h}{(1-B)(1-AB)} - \frac{B^{u_0+1}}{(1-B)(1-AB^2)} + \frac{B^{u_0+1}\sigma}{(1-B\sigma)(1-AB^2)}.$$

The first two terms are the sum of finite inner strips; they do not require $B < 1$. The convergence requirements are

$$(7.19) \quad AB < 1, \quad AB^2 < 1, \quad B\sigma < 1.$$

At $B = 1$ the removable expression is read by its limit, although none of the evaluations below has $B = 1$. Let $\mathcal{H}_{u_0 v_0}^{>P}$ be this rational expression minus the finite sum over $2b + v_0 \leq P$, $0 \leq a \leq b - h$. This is exactly the omitted pivot sum, not an asymptotic approximation.

The short multiplicities $p + 1 \leq m \leq 2p$, summed in (7.9), contribute for $p > P$ at most

$$(7.20) \quad \frac{z^8 s_4 L^{-2}}{\eta(L-1)} \sum_{u_0, v_0} \left\{ L^{2v_0} \mathcal{H}_{u_0 v_0}^{>P}(\eta, \eta L^4) - L^{v_0} \mathcal{H}_{u_0 v_0}^{>P}(\eta, \eta L^2) \right\}.$$

Indeed the multiplicity sum of L^{m-3} is $L^{-2}(L^{2p} - L^p)/(L-1)$, and $\eta^{e_D} \leq \eta^{a+b-1}$.

For the mixed tail put

$$t = \frac{252061464497}{250000000000}, \quad t^3 \geq T, \quad c_0 = (39/100)^2,$$

and define the rational function, on its convergence domain,

$$(7.21) \quad G_P(x) = \sum_{u_0, v_0} L^{2v_0} x^{v_0 - u_0} \mathcal{H}_{u_0 v_0}^{>P}(\eta/x^2, \eta L^4 x^2).$$

This is the nonnegative sum over omitted pivots of $L^{2p} \eta^{a+b} \sigma^{e_E} x^d$. Pair the bases $s = 2j+1, 2j+2$, $0 \leq j < d$. Their normalization leaves $w^{s-1-((s-1) \bmod 2)}$, so their envelope is $(1+L)L^{2p-2}J^j$. Thus the mixed multiplicity factor is

$$H_d = \frac{1 - J^d}{(1 - J)(1 - c_0 J^d)}.$$

The following elementary bound retains the numerator that would be lost by replacing its geometric sum with d .

Lemma 7.4. *For $0 < J, c_0 < 1$, $a_d \geq 0$ and $G(x) = \sum_{d \geq 1} a_d x^d < \infty$,*

$$\sum_{d \geq 1} a_d x^d H_d \leq \frac{G(x) - G(xJ) + \frac{c_0}{1 - c_0} \{G(xJ) - G(xJ^2)\}}{1 - J}.$$

Proof. For $X = J^d$, use $1/(1 - c_0 X) = 1 + c_0 X/(1 - c_0 X) \leq 1 + c_0 X/(1 - c_0)$, multiply by $(1 - X)/(1 - J)$ and sum. Both differences are sums of nonnegative terms of convergent series. $\square$

Since $T^{1+[d/3]} \leq Tt^d$, the mixed omitted tail is at most

$$(7.22) \quad \frac{C_0}{1 - J} \left[G_P(t) - G_P(tJ) + \frac{c_0}{1 - c_0} \{G_P(tJ) - G_P(tJ^2)\} \right], \quad C_0 = \frac{z^8 s_4 (1 + L) L^{-2} T}{\eta}.$$

All parameters here are rational. For $x = t, tJ, tJ^2$ the conditions (7.19) hold: passing from x to xJ leaves AB unchanged and decreases AB^2 and $B\sigma$. The initial inequalities are checked with the exact w of (7.11), rather than replacing w^2 by the slightly smaller β .

For the pure omitted sector, return to the exact factor β^N in (7.10). Equations (7.5) and (7.9) give, for every $m > 2p$,

$$T^{f_d} \beta^N P_{m,r,p} \leq T^{f_d} \beta^N A_{m,r,p}^{\text{abs}} \leq T^{f_d} \beta^N \mathcal{E}_{m,r,p}.$$

This argument does not require $P_{m,r,p} \leq Z_{m,r,p}$, which need not hold for a signed spectral expansion. To bound the sum, add the nonnegative terms $p < m \leq 2p$ and use $\beta^{\lfloor (m-1)/2 \rfloor - p}$ throughout. On the added range its exponent is nonpositive; on the original range the factor equals β^N . Put $J_0 = L^2\beta < 1$,

$$A_0 = \eta/t^2, \quad B_0 = \eta J_0 \beta^{-2} t^2, \quad M_1 = L^{-2} J_0 + L^{-1}, \quad M_2 = (L^{-2} + L^{-1}) J_0.$$

An upper bound for this enlarged sum with $p > P$ is

$$(7.23) \quad z^8 s_4 T \sum_{u_0, v_0} \frac{\beta^{-v_0} t^{v_0 - u_0} M_{v_0}}{\eta(1 - J_0)} \mathcal{H}_{u_0 v_0}^{>P}(A_0, B_0).$$

For $p = 2b + v_0$ the exact multiplicity sum needed here is

$$\sum_{m \geq p+1} L^{m-3} \beta^{\lfloor (m-1)/2 \rfloor - p} = \beta^{-v_0} (J_0 \beta^{-2})^b \frac{M_{v_0}}{1 - J_0}.$$

Substitute the pivot bounds to obtain (7.23). Again $J_0 < 1$ and all three conditions in (7.19) hold. The short-tail parameters in (7.20) also satisfy those three conditions. Thus none of these expressions subtracts divergent infinite sums.

7.5. The finite interval calculation. For the finite pivot prefixes the preceding formulas require finitely many exact graph evaluations. For each fixed graph, path dynamic programming sums over the endpoint spin, and a cycle is evaluated by fixing its initial spin and summing the closing edge. Loop restrictions are applied at the corresponding vertex. For pure and absolute-mode partitions the same multiplication is performed on the projected blocks. An upper enclosure of $\hat{A}_{m,r,p}^{\text{abs}}$ minus a lower enclosure of $\hat{P}_{m,r,p}$ is a valid upper enclosure of $\hat{M}_{m,r,p}$.

All interval endpoints are integers in units of 2^{-56} . Each multiplication and signed division is rounded outwards. The rational w , a quadratic enclosure for λ , and a rational enclosure for T are checked before evaluation. Integer endpoints are guarded by 2^{120} ; products that exceed signed 128-bit intermediate capacity are evaluated with arbitrary-precision integers. The resulting upper enclosures of the sums (7.15)–(7.17) are as follows. Here $U = 2^{56}$. The input enclosures are

$$\frac{73338785529844359}{U} \leq \lambda \leq \frac{73338785529844360}{U},$$

$$\frac{73854862990399022}{U} \leq T \leq \frac{73854862990399023}{U}.$$

The stored output fields `pure_hi_units`, `short_hi_units`, and `mixed_hi_units` are integer upper endpoints in units of U^{-1} ; the table displays their values after division by U .

TABLE 3. Finite pivot sums at $z = 1247/2000$ with complete multiplicity bounds. The integer numerators are scaled by $U = 2^{56}$.

Domain and evaluated quantity	Graph cases	Certified upper bound
P(400), two pure parity bases	159600	101235082714477743831/ U
S(100) and M(100), short and mixed bases	666600	S(100): 3738038979785516612/ U ; M(100): 36617965435406954/ U

The counts follow directly from $2 \sum_{p=2}^P (p-1) = P(P-1)$ and $\sum_{p=2}^P \sum_{r=1}^{p-1} (p+2(p-r)) = 2P(P+1)(P-1)/3$. The displayed bounds already include the respective geometric multiplicity denominators and feedback multipliers, not just the base-graph partitions. The finite graph evaluations establish the displayed interval bounds; the progression identities and inequalities above supply their extension to every remaining multiplicity for these pivots.

Combining the boxes with (7.20), (7.22) and (7.23) gives the following convenient outward ceilings. Each finite decimal in this table denotes the indicated rational number.

TABLE 4. Rational upper bounds for the six terms covering the cap-four mass. Every displayed decimal is a terminating rational value.

Contribution	Strict upper bound
Pure prefix, $p \leq 400$	1404.918996618068
Short prefix, $p \leq 100$	51.875711778803
Mixed prefix, $p \leq 100$	0.508176354267
Short omitted tail, $p > 100$	116.098136035634
Mixed omitted tail, $p > 100$	446.302688776606
Full feedback tail, $p > 400$	3.745874303725

The sum of the displayed ceilings is below 2024. Direct addition before rounding gives the slightly sharper certified value

$$A(1247/2000) < 2023.449583867101 < 2024.$$

Every genuine word has a unique pair of rightmost maxima and is covered by the contracted population. In (7.10), the short part is covered by $S(100)$ and (7.20), the pure part by $P(400)$ and (7.23), and the mixed part by $M(100)$ and (7.22). All six bounds are finite. Monotone convergence applied to the nonnegative sums proves convergence and completes the proof of (7.1).

8. COMPENSATION AND THE FINAL GENUS COMPARISON

Lemma 5.3 combines the depth-four residual leaves with the nonleaf parents of the unique-five residual leaves into disjoint parts of the population counted by A . It is a positive-real weighted comparison, not a coefficientwise comparison of equal genera. Substituting (7.1) in (5.4) gives, for every g ,

$$(8.1) \quad |\mathcal{E}_g| < 2024z_4^{-g} + 113z_5^{-g}, \quad z_4 = 1247/2000, \quad z_5 = 633/1000.$$

Proposition 8.1. *For every source genus $g \geq 837$, $|\mathcal{E}_g| < L(g)$, where $L(g)$ is the unused-target capacity of Proposition 4.3.*

Proof. Put

$$\bar{q} = \frac{309017}{500000}, \quad s_* = \frac{223607}{100000}, \quad r_* = \frac{809}{500}.$$

The rational inequalities $\bar{q}^2 + \bar{q} > 1$, $s_*^2 > 5$ give $\bar{q} > \varphi^{-1}$ and $s_* > \sqrt{5}$. Binet's formula therefore implies

$$\frac{|\mathcal{E}_g|}{F_g} < \Xi(g) := \frac{s_* \{2024(\bar{q}/z_4)^g + 113(\bar{q}/z_5)^g\}}{1 - \bar{q}^{2g}}.$$

The denominator uses the smaller $1 - \varphi^{-2g}$ for both parities; replacing φ^{-1} by $\bar{q}$ only enlarges the upper bound. Proposition 4.3 gives, throughout this range,

$$\frac{L(g)}{F_g} \geq C(g) := 1 + \frac{927}{500} - \frac{50r_*^2}{((63/100)r_*)^g}.$$

Exact rational evaluation at the joining genus gives

$$\Xi(837) < 2.850575297189 < 2.853985748250 < C(837).$$

Since $\bar{q} < z_4, z_5$ and $(63/100)r_* > 1$, the numerator terms of $\Xi(g)$ strictly decrease, its denominator increases, and $C(g)$ increases. The comparison follows for every $g \geq 837$. Failure of this particular scalar comparison at 836 has no implication for monotonicity at that genus; it is covered by the finite comparison. $\square$

9. THE FINITE COMPARISON

We describe the exact finite calculation used with the preceding estimates. All coefficients in this section are integers. The two pivots are the rightmost occurrences of the maximum coordinate; this choice prevents multiple counting when the pivot polynomials are summed.

Let $C_g(300)$ be the coefficient of z^g in the sum of the raw depth-four polynomials with rightmost pivot $p \leq 300$, over all multiplicities and left pivots. Set

$$(9.1) \quad U_g = C_g(300) + \mathbf{1}_{\{g \geq 604\}} \left\lfloor \frac{221715}{10^6} F_g \right\rfloor.$$

The omitted-pivot bound in Theorem 6.2 makes U_g an upper bound for the complete raw depth-four count. The support inequality for cap a ,

$$(9.2) \quad 2g \geq (a-1)m + p + 1,$$

also shows that $p > 300$ cannot contribute below genus 604 when $a = 4$. The floor is legitimate because the omitted coefficient is an integer. Notice that this term first enters U_g at $g = 604$ and first enters the shifted parent term U_{g-1} at $g = 605$.

For depth five, write $C_g^{(5)}$ for the coefficient of the complete raw two-pivot population and define

$$(9.3) \quad V_g = \begin{cases} C_g^{(5)}, & 0 \leq g \leq 300, \\ \lfloor 113(1000/633)^g \rfloor, & g \geq 301. \end{cases}$$

The first line is an exact coefficient calculation. The second line bounds the genuine multiple-five population by the weighted mass estimate of the preceding section; it is not a weighted-mass claim for the larger raw population. Coefficients with negative indices are zero.

Let N_g denote the no-one coefficients defined in the construction of the partial injection. The residual bound is

$$(9.4) \quad E_g^+ = U_g - N_g - N_{g-3} - N_{g-4} + U_{g-1} - N_{g-1} + V_g.$$

The shifted parent term has no sole-final-one subtraction: such a parent is permitted by the unique-five map.

Lemma 9.1 (Exact finite comparison). *For every $1 \leq g \leq 836$, the integers defined above satisfy $0 \leq E_g^+ \leq L(g)$.*

Verification and coverage. For a truncation degree H , inequality (9.2) bounds every contributing multiplicity. The retained depth-four calculation has $H = 897$, $p \leq 300$, and

$$3 \leq m \leq 597, \quad 2 \leq p \leq \min(300, m-1), \quad 1 \leq r < p.$$

These ranges comprise 17 775 550 graph triples in 595 disjoint multiplicity classes. Each component polynomial is computed by the path or cycle recurrence specified above, and the resulting products are summed and truncated to give $C_g(300)$.

For depth five with $H = 300$, the support inequality gives $m \leq 149$. The sufficient complete run over $3 \leq m \leq 149$ and all pivot pairs visits

$$\sum_{m=3}^{149} \binom{m-1}{2} = \binom{149}{3} = 540\,274$$

graph triples. The resulting exact array contains all 301 coefficients through degree 300. Thus no cap-five omitted-pivot estimate is used in the exact range of (9.3).

The rational series defining N_g and $L(g)$ are expanded by exact coefficient recurrences. Substitution into (9.1)–(9.4) checks all 836 inequalities. Put $\Delta_g = L(g) - E_g^+$. The smallest integer slack is 1, at $g = 1, 2$. Among $1 \leq g \leq 836$, the smallest Fibonacci-normalized slack is

$$\min_g \frac{\Delta_g}{F_g} = \frac{15028705}{33489287} > 0.448761,$$

attained at $g = 42$. Table 5 records the switches used by the proof. The displayed decimal entries are downward truncations of exact rational values. The capacity-normalized slack also satisfies

$$\frac{\Delta_g}{L(g)} > 0.179861 \quad (1 \leq g \leq 836),$$

with its minimum attained at $g = 605$.

TABLE 5. Transition checks for the finite comparison.

g	depth-five term	depth-four tail status	Δ_g/F_g
300	exact $C_{300}^{(5)}$	absent	> 0.871878
301	analytic mass bound	absent	> 0.683638
603	analytic mass bound	absent	> 0.872080
604	analytic mass bound	enters U_g	> 0.650368
605	analytic mass bound	enters U_{g-1}	> 0.513344
836	analytic mass bound	both terms present	> 0.513474

For completeness, the polynomial calculation introduces no numerical approximation. A polynomial truncated at degree H is evaluated at the digit base $b = 2^{H+2}$. The coefficient at actual weight $h \leq H$ is at most the number of positive compositions of h , hence less than b . Components are placed in a fixed coordinate order, and the unique pivot choice preserves this interpretation on summation. Minimum-weight normalization changes only the position of a digit; a state is pruned only when its least possible completed weight exceeds H . A negative normalization shift is made only after an exact divisibility check. Discarded higher digits therefore cannot alter a retained coefficient.

The exact transition rows, input identities, and regeneration procedure are recorded in the repository [11]. Proposition 5.2 and the preceding analytic bounds identify these integer comparisons with the required bound on $|\mathcal{E}_g|$. $\square$

At the other end of the finite range, $\Delta_{836}/F_{836} > 0.513474$. The scalar analytic comparison of Proposition 8.1 fails at 836, but at 837 its exact rational margin satisfies

$$C(837) - \Xi(837) > 0.0034104510, \quad \frac{C(837) - \Xi(837)}{C(837)} > 0.0011949783.$$

Thus the finite and analytic arguments meet without an uncovered genus.

Proof of Theorem 1.1. For $g = 0$, the two counts are $n_0 = n_1 = 1$. For $1 \leq g \leq 836$, the residual bound and Lemma 9.1 give $|\mathcal{E}_g| \leq L(g)$. For $g \geq 837$, Proposition 8.1 gives the same comparison. Applying (5.2) yields the result. $\square$

10. CONCLUDING REMARKS

The stronger Fibonacci inequality remains open:

$$n_{g+1} \geq n_g + n_{g-1} \quad (g \geq 1).$$

The present construction guarantees enough unused targets to accommodate its exceptional sources, but does not provide the additional n_{g-1} targets required for this inequality. The finer error terms studied by Zhu [14] offer a complementary way to formulate that obstruction.

It would also be useful to replace the comparison of cardinalities for the exceptional sources by a direct operation on their Kunz words, with an explicit inverse on its image. The theorem already guarantees an injection between the two finite genus classes. The question is whether the remaining matching can be explained by a structural rule, rather than by estimates and finite verification.

AI STATEMENT

This work was developed through continuous collaboration between the author and AI models. GPT-6 Astra (OpenAI) and Fable 5.1 (Anthropic) contributed to proof development and checking, exact computation, literature research, and preparation and review of the manuscript. The author takes full responsibility for all mathematical results and the contents of this manuscript.

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APPENDIX A. COMPUTATIONAL MATERIAL

The computational part has three distinct tasks: generating the raw coefficient arrays, bounding the finite weighted graph sums, and checking the resulting scalar inequalities. Table 6 records their parameter ranges. The verification of stored outputs and the regeneration of those outputs are separate procedures.

TABLE 6. Finite inputs to the proof. The retained depth-four arrays cover a wider range than the theorem requires.

Calculation	Range	Graphs
Raw depth-four coefficients	$p \leq 300, 3 \leq m \leq 597$; degrees 0–897	17 775 550
Raw depth-five coefficients	$3 \leq m \leq 149$; all pivots; degrees 0–300	540 274
Pure-mode weighted sum	$p \leq 400$	159 600
Short and mixed weighted sums	$p \leq 100$	666 600

The depth-five row describes sufficient regeneration coverage. The retained file was originally produced over $3 \leq m \leq 293$ (4 149 466 parameter triples); the support inequality makes all additional contributions zero through degree 300. Regeneration over $3 \leq m \leq 149$ gives the same 301 coefficients.

The coefficient calculation uses arbitrary-precision integer arithmetic. The weighted graph sums use rational enclosures with outward rounding, and the final scalar comparisons use exact rational arithmetic. The calculation of an enclosure, its propagation through a recurrence, and the final strict comparison are all specified by the mathematical bounds in the text.

Code and data availability. The source code and exact finite outputs are available in the public repository, version v1 [11]. Its verification guide maps the computations in the text to their inputs and commands, with separate instructions for checking stored results and regenerating coefficient arrays and interval bounds. All Python checks use the standard library; regeneration requires a C++17 compiler with 128-bit integer support and the GMP library.

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